

# 04 — Muscle Force Generation and Motor Control

## BMED365 – Computational Medicine (Lab 5)

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Updated: 2026-02-19 (Cursor 2.5.17 with Claude Opus 4.6)

Updated: 2026-02-20 (Cursor 2.5.20 with Claude Opus 4.6) - saved notebook to .pdf

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## 1. Motivation & Learning Objectives

### Why study muscle force generation and motor control?

Every voluntary movement — from the blink of an eye to the swing of a tennis racket — originates in an exquisitely orchestrated chain of events: the brain formulates a motor plan, spinal motor neurons fire action potentials, and skeletal muscles contract by converting electrochemical signals into mechanical force. **Understanding this chain is central to biomedicine, rehabilitation, sports science, and robotics.**

Muscle is the body's engine. It accounts for roughly **40% of total body mass** and is the sole effector through which the nervous system interacts with the physical world. Despite its everyday familiarity, the quantitative understanding of muscle contraction required breakthroughs spanning more than a century — from A.V. Hill's thermodynamic experiments in the 1920s, through the discovery of the sliding filament mechanism in 1954, to modern computational models that can predict force output from neural drive.

Computational models of muscle and motor control are now indispensable tools in:

- **Clinical medicine** — understanding neuromuscular diseases (e.g. ALS, muscular dystrophy, spasticity after stroke)
- **Rehabilitation engineering** — designing functional electrical stimulation (FES) protocols for paralyzed limbs
- **Prosthetics and orthotics** — creating powered exoskeletons and myoelectric prostheses
- **Sports science** — optimizing training regimes and predicting injury risk
- **Robotics** — building biologically inspired actuators and controllers for humanoid robots
- **Surgical planning** — simulating musculoskeletal surgery outcomes (e.g. tendon transfer, joint replacement)

### What you will learn

In this notebook you will:

- 1. **Understand** the hierarchical structure of skeletal muscle — from whole muscle down to the molecular machinery of actin and myosin
- 2. **Appreciate** the historical context — how Hill, Huxley, and others built our quantitative understanding of contraction
- 3. **Master** the key mathematical relationships: force-length, force-velocity, and activation dynamics
- 4. **Implement** a Hill-type muscle model in Python and simulate isometric and isotonic contractions
- 5. **Explore** motor unit recruitment and the size principle (Henneman's principle)
- 6. **Simulate** motor control — from neural command signals to limb movement
- 7. **Connect** human muscle physiology to robotic actuator design and control
- 8. **Examine** clinical implications — from muscle fatigue to neuromuscular disease

## 2. Historical Context: From Frogs' Legs to Molecular Motors

### A timeline of discovery

The science of muscle contraction has a rich history intertwined with some of the most fundamental advances in physiology, biophysics, and biomechanics.

| Year  | Milestone                                                                  | Scientists                                                                         |
|-------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 1791  | "Animal electricity" — frog leg twitches from electrical stimulation       | Luigi <b>Galvani</b>                                                               |
| 1840s | First measurements of heat produced during muscle contraction              | Hermann von <b>Helmholtz</b>                                                       |
| 1922  | Thermodynamic analysis of muscle: heat = activation heat + shortening heat | Archibald Vivian <b>Hill</b>                                                       |
| 1938  | <b>Hill's equation</b> — the hyperbolic force-velocity relationship        | A.V. <b>Hill</b> (Nobel Prize 1922)                                                |
| 1949  | Discovery of the sarcomere's banding pattern by electron microscopy        | H.E. <b>Huxley</b>                                                                 |
| 1953  | X-ray diffraction studies of muscle structure                              | Hugh <b>Huxley</b>                                                                 |
| 1954  | <b>Sliding filament theory</b> — simultaneously and independently proposed | H.E. <b>Huxley</b> & J. <b>Hanson</b> ; A.F. <b>Huxley</b> & R. <b>Niedergerke</b> |
| 1957  | Cross-bridge theory — molecular model of force generation                  | A.F. <b>Huxley</b>                                                                 |
| 1965  | <b>Size principle</b> of motor unit recruitment                            | Elwood <b>Henneman</b>                                                             |
| 1969  | Refined Hill-type muscle model with activation dynamics                    | Felix <b>Zajac</b> (later, 1989)                                                   |
| 1971  | Swinging cross-bridge model with biochemical states                        | A.F. Huxley & R.M. <b>Simmons</b>                                                  |

| Year  | Milestone                                                 | Scientists               |
|-------|-----------------------------------------------------------|--------------------------|
| 2000s | Large-scale musculoskeletal simulations (OpenSim)         | Scott <b>Delp</b> et al. |
| 2010s | Bioinspired robotic actuators mimicking muscle properties | Various groups           |

## The two Huxleys

A common source of confusion: **Hugh Erskine Huxley** (1924–2013) and **Sir Andrew Fielding Huxley** (1917–2012) were *not related*. Both made independent, complementary contributions to our understanding of muscle:

- **H.E. Huxley** used electron microscopy and X-ray diffraction to reveal the structural basis of contraction
- **A.F. Huxley** (who also co-authored the Hodgkin-Huxley model of the action potential!) developed the mathematical cross-bridge theory

A.F. Huxley received the **Nobel Prize in Physiology or Medicine in 1963** (shared with Hodgkin and Eccles) for work on nerve impulses — but his muscle work was equally groundbreaking.

## A.V. Hill and the force-velocity relationship

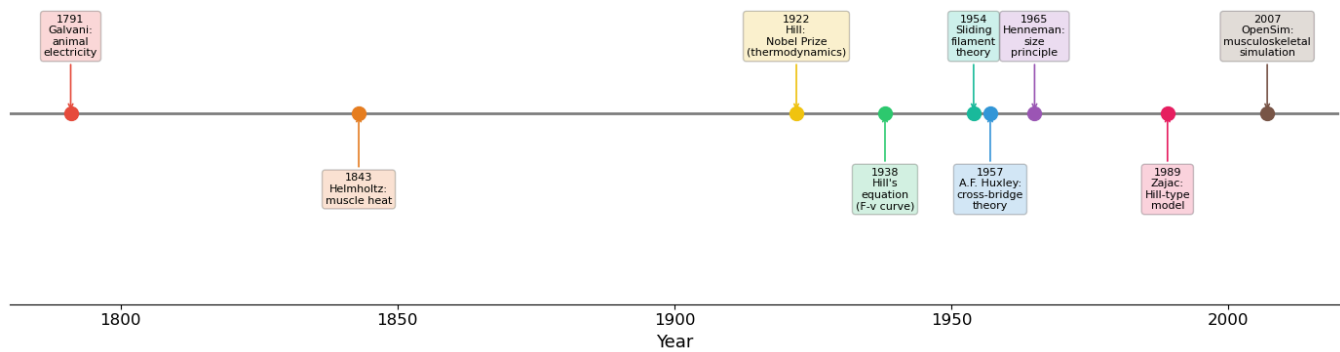
**Archibald Vivian Hill** (1886–1977) was a British physiologist who received the **Nobel Prize in Physiology or Medicine in 1922** for discoveries relating to heat production in muscle. His most lasting contribution to muscle mechanics came in 1938, when he published the **force-velocity equation** — a simple hyperbolic relationship that describes how the speed of muscle shortening depends on the load:

$$(F + a)(v + b) = (F_0 + a) \cdot b$$

where  $F$  is the muscle force,  $v$  is the shortening velocity,  $F_0$  is the maximum isometric force, and  $a$ ,  $b$  are constants related to the muscle's thermodynamic properties. This remarkably simple equation remains the foundation of virtually all computational muscle models used today.

*"The combination of observation and of theory, which is necessary if physiology is to develop as a science, requires at every point that theory should be tested by observation, and observation should be guided by theory." — A.V. Hill*

Figure 1. Key Milestones in Muscle Physiology and Motor Control



## 3. Muscle Structure: From Whole Muscle to Molecular Motors

### The hierarchical organization of skeletal muscle

Skeletal muscle has a beautiful hierarchical structure, with each level of organization serving a specific mechanical and physiological function:

| Level                      | Structure                       | Size                                             | Function                             |
|----------------------------|---------------------------------|--------------------------------------------------|--------------------------------------|
| <b>Whole muscle</b>        | e.g. biceps brachii             | ~10–30 cm                                        | Generates movement across joints     |
| <b>Fascicle</b>            | Bundle of fibers                | ~1–10 mm diameter                                | Organized force transmission         |
| <b>Muscle fiber (cell)</b> | Multinucleated cell             | ~10–100 $\mu\text{m}$ diameter, up to 30 cm long | The functional contractile unit      |
| <b>Myofibril</b>           | Cylindrical organelle           | ~1–2 $\mu\text{m}$ diameter                      | Contains the contractile machinery   |
| <b>Sarcomere</b>           | Repeating unit of myofibril     | ~2.0–2.5 $\mu\text{m}$ at rest                   | The fundamental contractile unit     |
| <b>Myofilaments</b>        | Actin (thin) and myosin (thick) | ~5–15 nm diameter                                | Molecular motors that generate force |

### Muscle fiber types

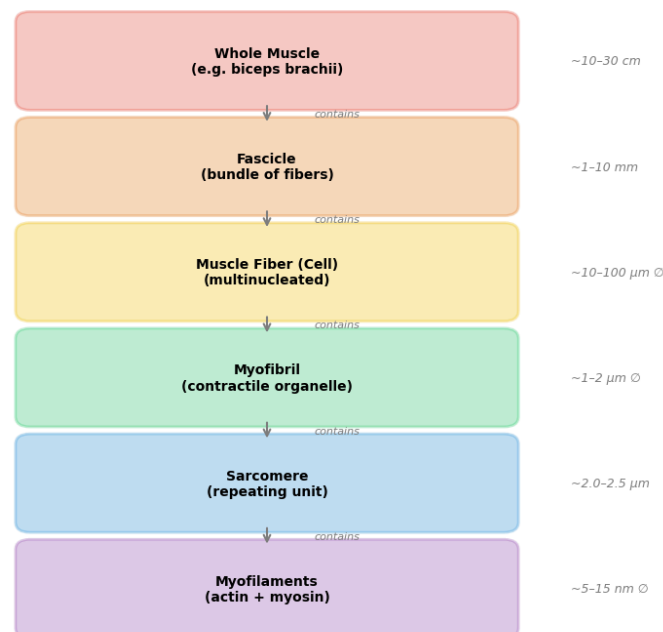
Not all muscle fibers are the same. Human skeletal muscle contains a mix of fiber types that differ in their contractile speed, fatigue resistance, and metabolic properties:

| Property                  | Type I (Slow oxidative) | Type IIa (Fast oxidative-glycolytic) | Type IIx (Fast glycolytic) |
|---------------------------|-------------------------|--------------------------------------|----------------------------|
| <b>Contraction speed</b>  | Slow                    | Fast                                 | Fastest                    |
| <b>Fatigue resistance</b> | Very high               | Moderate                             | Low                        |

| Property              | Type I (Slow oxidative) | Type IIa (Fast oxidative-glycolytic) | Type IIx (Fast glycolytic) |
|-----------------------|-------------------------|--------------------------------------|----------------------------|
| Force production      | Low                     | Moderate                             | High                       |
| Mitochondrial density | High                    | Moderate-High                        | Low                        |
| Color                 | Red (myoglobin-rich)    | Red                                  | White                      |
| Primary use           | Posture, endurance      | Mixed activities                     | Explosive movements        |
| Example               | Soleus (standing)       | Gastrocnemius                        | Quadriceps (sprinting)     |

The mix of fiber types varies between muscles and between individuals — elite marathon runners have a higher proportion of Type I fibers, while sprinters have more Type II fibers.

**Figure 2. Hierarchical Organization of Skeletal Muscle**



## 4. The Sliding Filament Theory and the Cross-Bridge Cycle

### The sliding filament theory (1954)

The **sliding filament theory** is one of the most important discoveries in biology. Proposed independently in 1954 by two pairs of scientists — **H.E. Huxley & Jean Hanson** and **A.F. Huxley & Rolf Niedergerke** — it states that:

**Muscle contraction occurs because the thin (actin) and thick (myosin) filaments slide past each other, without the filaments themselves changing length.**

This was revolutionary because earlier theories assumed that the protein filaments shortened like coiling springs. The sliding filament theory explained why:

- The **A-band** (region of thick filaments) stays constant in width during contraction
- The **I-band** (region of thin filaments only) narrows
- The **H-zone** (region of thick filaments only) narrows
- The overall **sarcomere** shortens

## The sarcomere: the fundamental contractile unit

The sarcomere is bounded by **Z-discs** (or Z-lines) and contains:

- **Thin filaments (actin)**: anchored at the Z-disc, extend toward the center
- **Thick filaments (myosin)**: centered in the sarcomere, with globular "heads" that can bind to actin
- **Titin**: a giant elastic protein connecting the thick filament to the Z-disc, providing passive stiffness
- **Regulatory proteins**: troponin and tropomyosin on actin filaments, which control access to myosin binding sites

## The cross-bridge cycle

Force generation occurs through the **cross-bridge cycle** — a repeating sequence of molecular events powered by ATP hydrolysis:

1. **Attachment**: Myosin head binds to actin (forms a "cross-bridge")
2. **Power stroke**: Myosin head pivots, pulling the actin filament toward the center of the sarcomere (force generation!)
3. **Detachment**: ATP binds to myosin, causing it to release from actin
4. **Recovery stroke**: ATP is hydrolyzed ( $\rightarrow$  ADP + Pi), re-cocking the myosin head for the next cycle

Each cycle moves the actin filament by about **5–10 nm** and generates a force of about **2–4 pN** (piconewtons). A single muscle fiber contains billions of cross-bridges operating asynchronously, producing smooth, sustained force.

## Excitation-contraction coupling

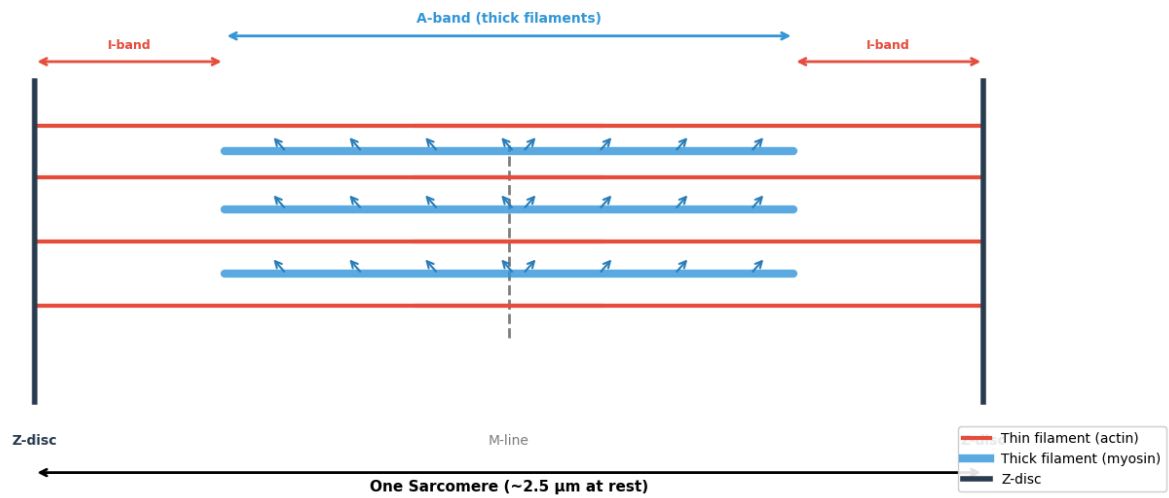
The link between the neural signal and contraction involves:

1. **Motor neuron** fires an action potential
2. Acetylcholine released at the **neuromuscular junction**
3. Action potential propagates along the muscle fiber membrane (sarcolemma) and into **T-tubules**
4.  $\text{Ca}^{2+}$  released from the **sarcoplasmic reticulum** (SR)
5.  $\text{Ca}^{2+}$  binds to **troponin**, causing **tropomyosin** to shift and expose myosin binding sites on actin

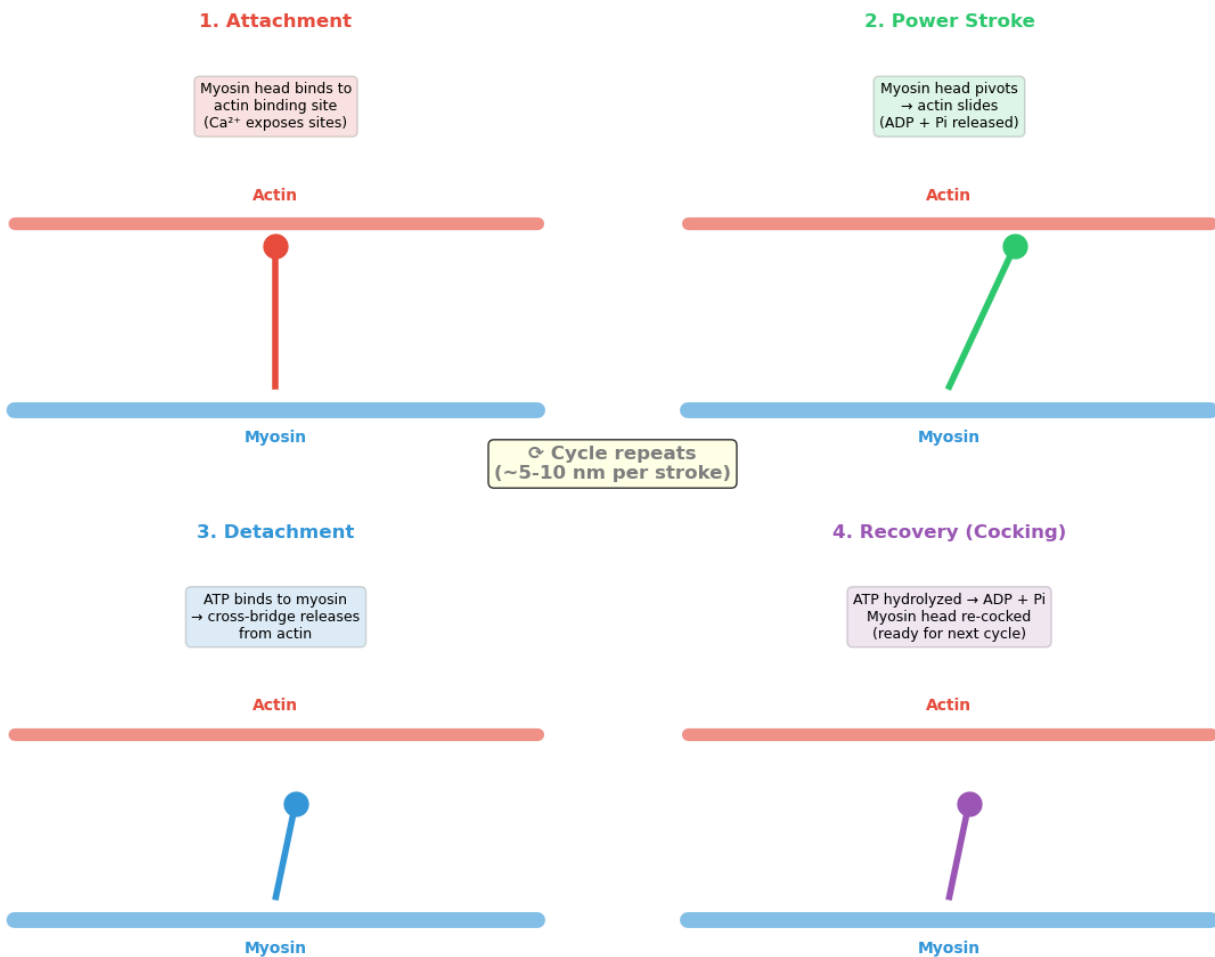
6. **Cross-bridge cycling** begins → force generation

7.  $\text{Ca}^{2+}$  pumped back into SR → relaxation

**Figure 3. Sarcomere Structure (Schematic)**

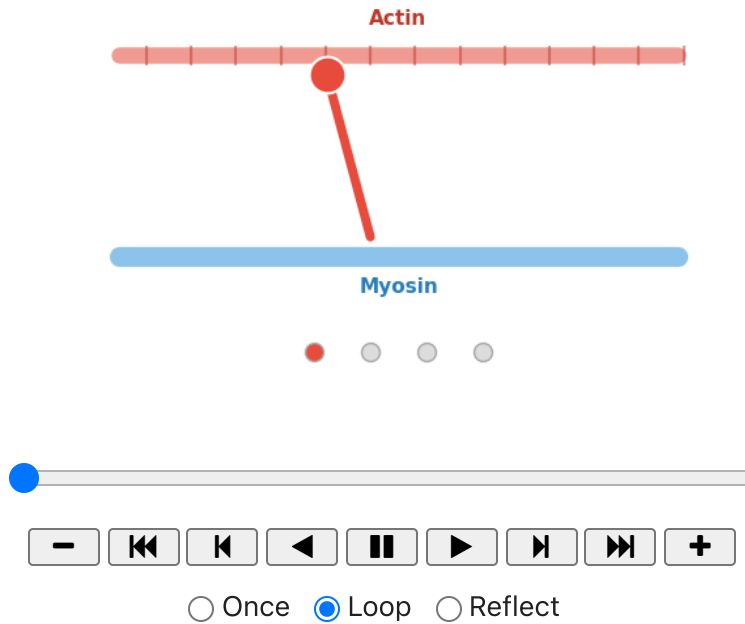


**Figure 4. The Cross-Bridge Cycle**



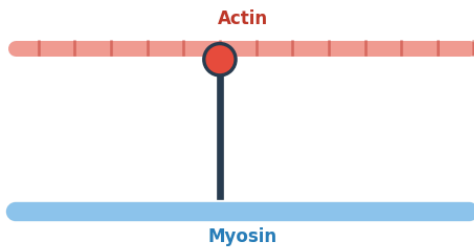
## 1. Attachment

$\text{Ca}^{2+}$  exposes binding sites  $\rightarrow$  myosin head binds to actin

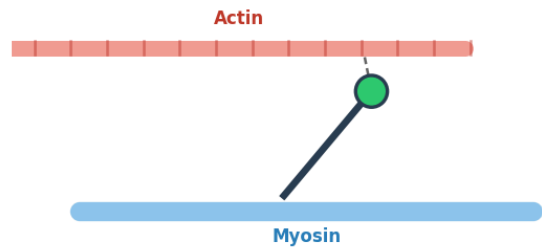


### Cross-Bridge Power Stroke Cycle — Static Snapshot (4 Phases)

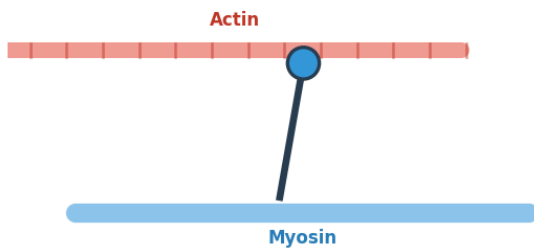
**1. Attachment**  
 $\text{Ca}^{2+}$  exposes binding sites  $\rightarrow$  myosin binds actin



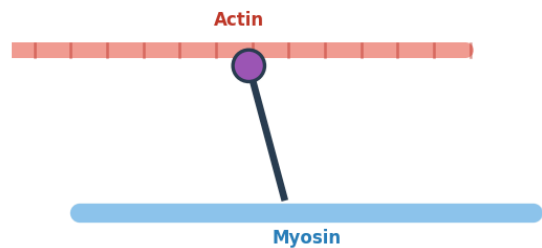
**2. Power Stroke**  
Myosin pivots, pulling actin (~10 nm)



**3. Detachment**  
ATP binds  $\rightarrow$  cross-bridge releases



**4. Recovery (Cocking)**  
ATP hydrolyzed  $\rightarrow$  head re-cocked





# Modeling Cross-Bridge Dynamics with ODEs

The four-phase cross-bridge cycle can be modeled as a **four-state chemical kinetic system**. Let  $x_i(t)$  denote the fraction of cross-bridges in state  $i$ :

| State | Variable | Description                                    |
|-------|----------|------------------------------------------------|
| 1     | $x_1$    | Detached, cocked (ready to bind)               |
| 2     | $x_2$    | Attached, pre-power-stroke                     |
| 3     | $x_3$    | Attached, post-power-stroke (force-generating) |
| 4     | $x_4$    | Detached, uncocked                             |

The transitions follow **first-order kinetics** with rate constants  $k_{ij}$  (units:  $\text{s}^{-1}$ ):

$$\frac{dx_1}{dt} = k_{41} x_4 - k_{12} ([\text{Ca}^{2+}]) x_1$$

$$\frac{dx_2}{dt} = k_{12} x_1 - k_{23} x_2$$

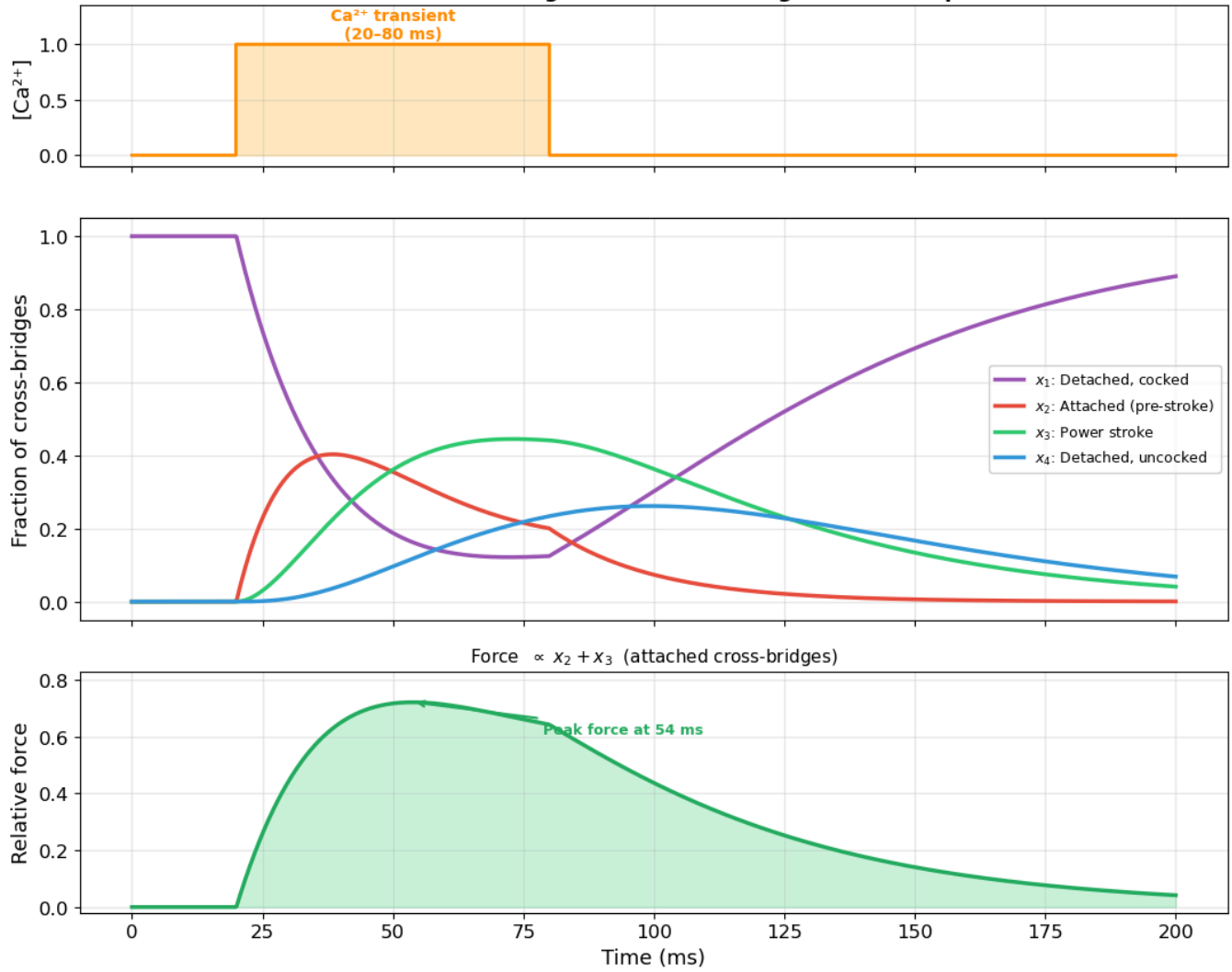
$$\frac{dx_3}{dt} = k_{23} x_2 - k_{34} x_3$$

$$\frac{dx_4}{dt} = k_{34} x_3 - k_{41} x_4$$

Key features:

- $k_{12}$  (attachment) depends on **Ca<sup>2+</sup> concentration** — binding sites are only exposed when Ca<sup>2+</sup> binds troponin
- $k_{34}$  (detachment) depends on **ATP availability** — ATP binding is needed to release the cross-bridge
- **Force**  $\propto x_2 + x_3$  (the fraction of attached cross-bridges)
- The total is conserved:  $x_1 + x_2 + x_3 + x_4 = 1$  (so only 3 of the 4 ODEs are independent)

### Four-State Cross-Bridge ODE Model — Single Twitch Response



Steady-state (full  $Ca^{2+}$ ):  $x_1=0.16$   $x_2=0.19$   $x_3=0.38$   $x_4=0.27$

Attached fraction ( $\propto$  force): 57.0%

Slowest step: detachment ( $k_{34} = 25 \text{ s}^{-1}$ ,  $\tau = 40 \text{ ms}$ )



**Q3.** Why does ATP need to bind to myosin for muscle *relaxation* (not just contraction)?

► Click to reveal answer

## 5. Scientific and Clinical Implications

### Why computational muscle models matter in medicine

Understanding muscle force generation quantitatively has profound implications across many areas of medicine and biomedical engineering:

#### Neuromuscular diseases

| Disease                           | What goes wrong                                                                   | How models help                                                         |
|-----------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>Muscular dystrophy</b>         | Progressive muscle fiber degeneration (e.g. dystrophin deficiency in Duchenne MD) | Predict force loss as fiber population changes; optimize rehabilitation |
| <b>ALS (Lou Gehrig's disease)</b> | Motor neuron death → progressive paralysis                                        | Model motor unit loss and remaining capacity; plan FES interventions    |
| <b>Myasthenia gravis</b>          | Antibodies block acetylcholine receptors at NMJ                                   | Simulate reduced neuromuscular transmission efficiency                  |
| <b>Spasticity (post-stroke)</b>   | Loss of inhibitory control → excessive muscle tone                                | Model altered recruitment patterns and co-contraction                   |
| <b>Sarcopenia</b>                 | Age-related loss of muscle mass and motor units                                   | Predict functional decline; design exercise interventions               |

#### Rehabilitation and prosthetics

- **Functional Electrical Stimulation (FES):** Computational models predict optimal stimulation patterns to produce desired movements in paralyzed limbs
- **Myoelectric prostheses:** EMG-based control algorithms use muscle models to decode intended movements from residual muscle signals
- **Exoskeletons:** Human-robot interaction models must account for the nonlinear force-length-velocity properties of muscle
- **Tendon transfer surgery:** Surgeons use musculoskeletal simulations (e.g. OpenSim) to predict functional outcomes of surgical procedures

#### Sports science and ergonomics

- **Injury prediction:** Models of muscle strain during eccentric contractions help identify injury-prone conditions
- **Performance optimization:** Simulations can identify which muscles limit performance in specific movements

- **Fatigue modeling:** Understanding how repeated contractions degrade force output
- **Workplace ergonomics:** Predicting muscle fatigue and injury risk from repetitive tasks

## Cardiac muscle

The same cross-bridge principles apply to cardiac muscle, where computational models are used to:

- Simulate heart contraction and predict cardiac output
- Understand cardiomyopathies at the molecular level
- Design and test cardiac assist devices

## 6. The Hill Muscle Model: Conceptual Framework

### From biology to engineering

To simulate muscle force computationally, we need a model that is biologically meaningful yet tractable enough to embed in larger simulations. The **Hill-type muscle model** (refined by Zajac, 1989) achieves this by representing the muscle-tendon unit as a combination of mechanical elements:

| Element                         | Symbol | Represents                           | Property                                          |
|---------------------------------|--------|--------------------------------------|---------------------------------------------------|
| <b>Contractile Element</b>      | CE     | Active cross-bridge force generation | Force depends on length, velocity, and activation |
| <b>Series Elastic Element</b>   | SEE    | Tendon + aponeurosis                 | Stores and releases elastic energy                |
| <b>Parallel Elastic Element</b> | PEE    | Passive connective tissue, titin     | Resists stretch at extreme lengths                |

### The key relationships

The Hill-type model captures three fundamental properties of muscle:

1. **Force-Length Relationship**  $f_L(\tilde{l})$ : Active force is maximal at an optimal sarcomere length and decreases at shorter and longer lengths (the "inverted U" curve)
2. **Force-Velocity Relationship**  $f_V(\tilde{v})$ :
  - During **shortening** (concentric): force *decreases* with increasing velocity (Hill's equation)
  - During **lengthening** (eccentric): force *increases* above isometric maximum (up to  $\sim 1.5\text{--}1.8 \times F_0$ )
3. **Activation Dynamics**  $a(t)$ : The time-dependent process of  $\text{Ca}^{2+}$  release and reuptake, converting the neural excitation  $u(t)$  into a muscle activation level  $a(t) \in [0, 1]$

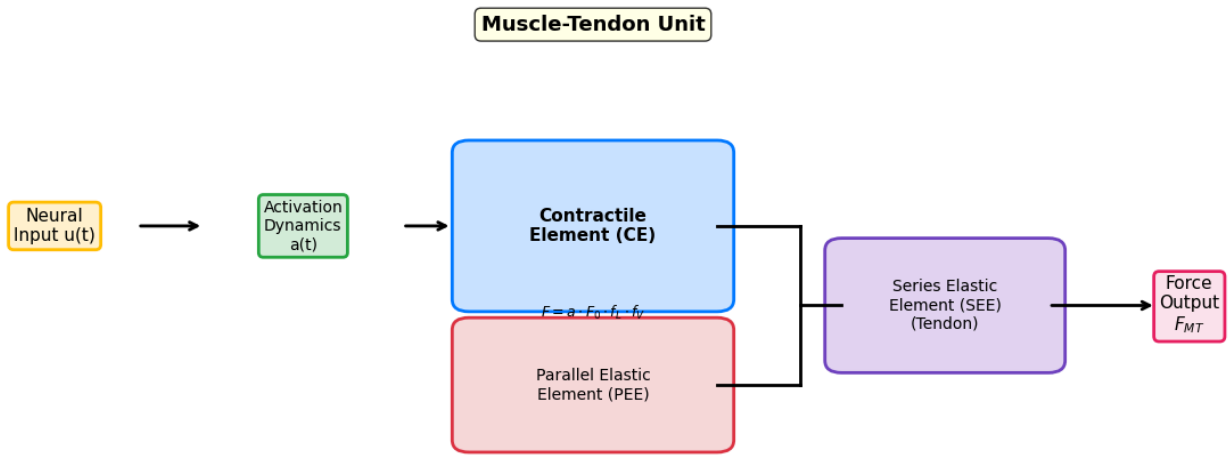
The total active muscle force is:

$$F_{\text{muscle}} = a(t) \cdot F_0 \cdot f_L(\tilde{l}) \cdot f_V(\tilde{v}) + F_{\text{passive}}(\tilde{l})$$

where:

- $a(t)$  is the activation level (0 = relaxed, 1 = fully activated)
- $F_0$  is the maximum isometric force at optimal length
- $f_L$  is the normalized force-length factor
- $f_V$  is the normalized force-velocity factor
- $\tilde{l} = l/l_0$  is the normalized muscle length
- $\tilde{v} = v/v_{\text{max}}$  is the normalized shortening velocity
- $F_{\text{passive}}$  is the passive elastic force

**Figure 5. Hill-Type Muscle-Tendon Unit Model**



## 7. Mathematical Formulation

### 7.1 Force-Length Relationship

The active force-length relationship arises directly from the sliding filament theory: force depends on the degree of overlap between actin and myosin filaments. At the **optimal length**  $l_0$ , overlap is maximal and the most cross-bridges can form. Force drops at shorter lengths (filament collision and interference) and at longer lengths (reduced overlap).

A commonly used Gaussian approximation is:

$$f_L(\tilde{l}) = \exp \left[ - \left( \frac{\tilde{l} - 1}{\gamma} \right)^2 \right]$$

where  $\tilde{l} = l/l_0$  is the normalized fiber length and  $\gamma \approx 0.45$  controls the width of the curve.

The **passive force** (from titin and connective tissue) increases exponentially at long lengths:

$$f_{\text{passive}}(\tilde{l}) = \begin{cases} k_{PE} \cdot (\tilde{l} - 1)^2 & \text{if } \tilde{l} > 1 \\ 0 & \text{if } \tilde{l} \leq 1 \end{cases}$$

## 7.2 Force-Velocity Relationship (Hill's Equation)

A.V. Hill's 1938 equation describes the hyperbolic relationship between force and shortening velocity. In normalized form:

**Shortening** ( $\tilde{v} < 0$ , concentric contraction):

$$f_V(\tilde{v}) = \frac{1 + \tilde{v}/\tilde{v}_{\max}}{1 - \tilde{v}/(\tilde{v}_{\max} \cdot A_f)}$$

where  $A_f \approx 0.25$  is a shape factor.

**Lengthening** ( $\tilde{v} > 0$ , eccentric contraction):

$$f_V(\tilde{v}) = \frac{F_{\text{ecc,max}} - (F_{\text{ecc,max}} - 1) \cdot \frac{1 + \tilde{v}/\tilde{v}_{\max}}{1 - \tilde{v} \cdot k_{\text{ecc}}/(\tilde{v}_{\max} \cdot A_f)}}{1}$$

For simplicity, we will use a piecewise rational function that captures the essential behavior.

## 7.3 Activation Dynamics

The neural excitation  $u(t) \in [0, 1]$  is transformed into muscle activation  $a(t) \in [0, 1]$  through first-order dynamics representing  $\text{Ca}^{2+}$  kinetics:

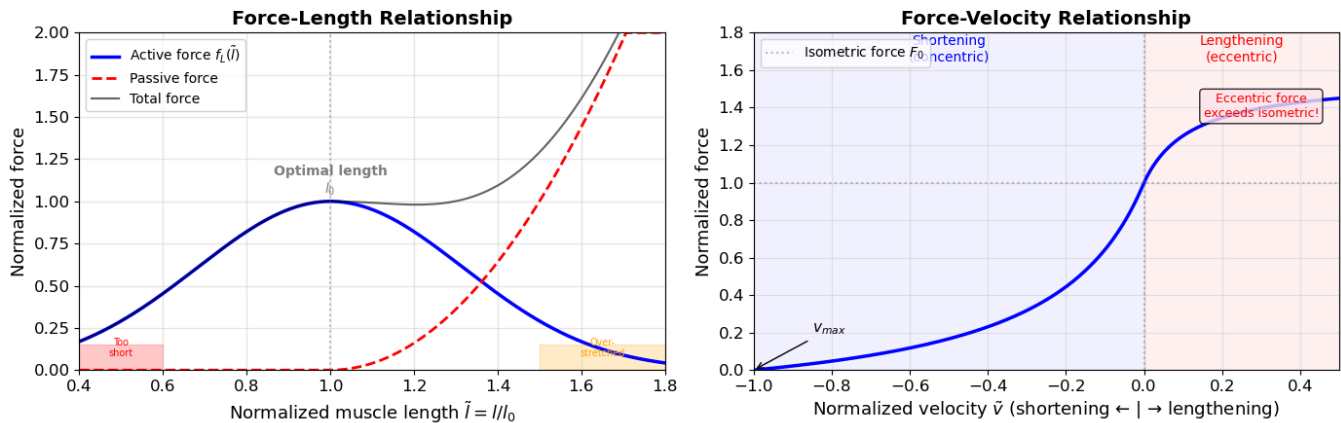
$$\frac{da}{dt} = \frac{u(t) - a(t)}{\tau(u, a)}$$

where the time constant differs for activation and deactivation:

$$\tau(u, a) = \begin{cases} \tau_{\text{act}} = 10\text{--}20 \text{ ms} & \text{if } u > a \text{ (activation)} \\ \tau_{\text{deact}} = 40\text{--}60 \text{ ms} & \text{if } u \leq a \text{ (deactivation)} \end{cases}$$

Activation is faster than deactivation because  $\text{Ca}^{2+}$  release from the SR is faster than reuptake — an asymmetry with important functional consequences for motor control.

Figure 6. The Two Fundamental Properties of Muscle



## Check Your Understanding: Force-Length and Force-Velocity

**Q4.** A patient has a muscle contracture (the muscle is chronically shortened). Based on the force-length relationship, would you expect this muscle to produce more or less active force than normal at its resting position?

► Click to reveal answer

**Q5.** During which type of contraction (concentric or eccentric) is a muscle most likely to be injured? Why?

► Click to reveal answer

**Q6.** Hill's force-velocity equation shows that a muscle generates zero force at its maximum shortening velocity  $v_{max}$ . What does this mean physically?

► Click to reveal answer

## 8. Motor Unit Recruitment and the Size Principle

### What is a motor unit?

A **motor unit** consists of a single **alpha motor neuron** (in the spinal cord) and all the muscle fibers it innervates. It is the fundamental unit of motor control — the smallest unit of force that the nervous system can independently control.

Key properties:

- A single motor neuron may innervate **5 to 2000+** muscle fibers
- All fibers in a motor unit are of the **same type** (Type I, IIa, or IIx)
- All fibers in a motor unit contract **simultaneously** when the motor neuron fires



- The force of a motor unit depends on its **size** (number of fibers) and **firing rate**

## Henneman's size principle (1965)

Elwood Henneman discovered that motor units are recruited in a **fixed order**, from smallest to largest:

1. **Small motor units** (Type I, slow, fatigue-resistant) are recruited first for low-force tasks
2. **Medium motor units** (Type IIa) are added as force demand increases
3. **Large motor units** (Type IIx, fast, fatigable) are recruited last, only for high-force or explosive tasks

This orderly recruitment has elegant functional consequences:

- **Fine control at low forces:** Small increments in force for delicate tasks (e.g. threading a needle)
- **Power when needed:** Large, fast motor units available for emergencies (e.g. catching yourself from falling)
- **Fatigue resistance:** The most fatigue-resistant units are used most, preserving endurance

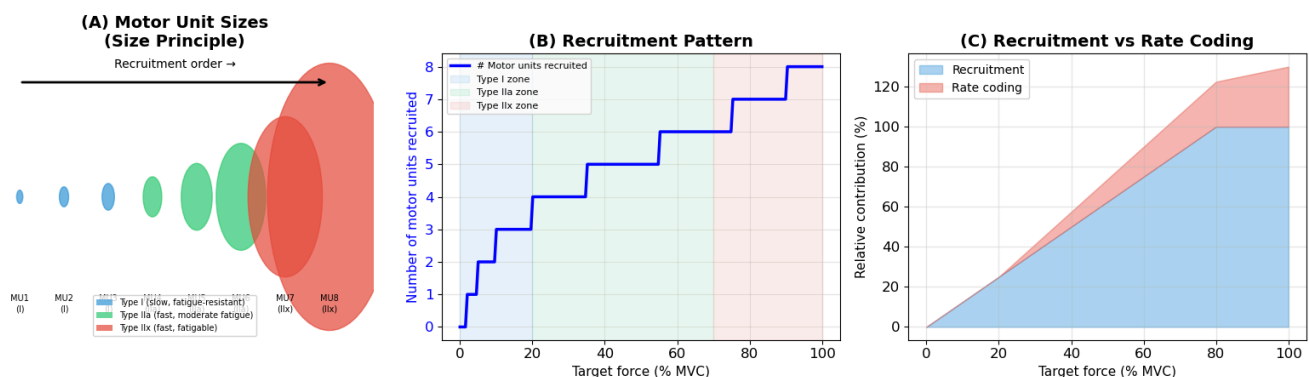
## Force modulation: two mechanisms

The nervous system controls muscle force through two complementary mechanisms:

1. **Recruitment:** Activating additional motor units (dominant at low force levels, < 50–80% of maximum)
2. **Rate coding:** Increasing the firing rate of already-active motor units (dominant at high force levels)

The combination of recruitment and rate coding gives the motor system an enormous dynamic range — from the few millinewtons needed to hold a feather to the thousands of newtons generated during a maximal jump.

**Figure 7. Motor Unit Recruitment and the Size Principle**



## DIY Exercise: Motor Unit Arithmetic

**Exercise 1.** A muscle has 100 motor units with the following distribution:

- 60 Type I units, each producing 0.5 N
- 30 Type IIa units, each producing 2.0 N
- 10 Type IIx units, each producing 8.0 N

What is the maximum force the muscle can produce?

► Click to reveal answer

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**Exercise 2.** If a person with ALS loses their 10 largest motor units first, what percentage of maximum force is lost?

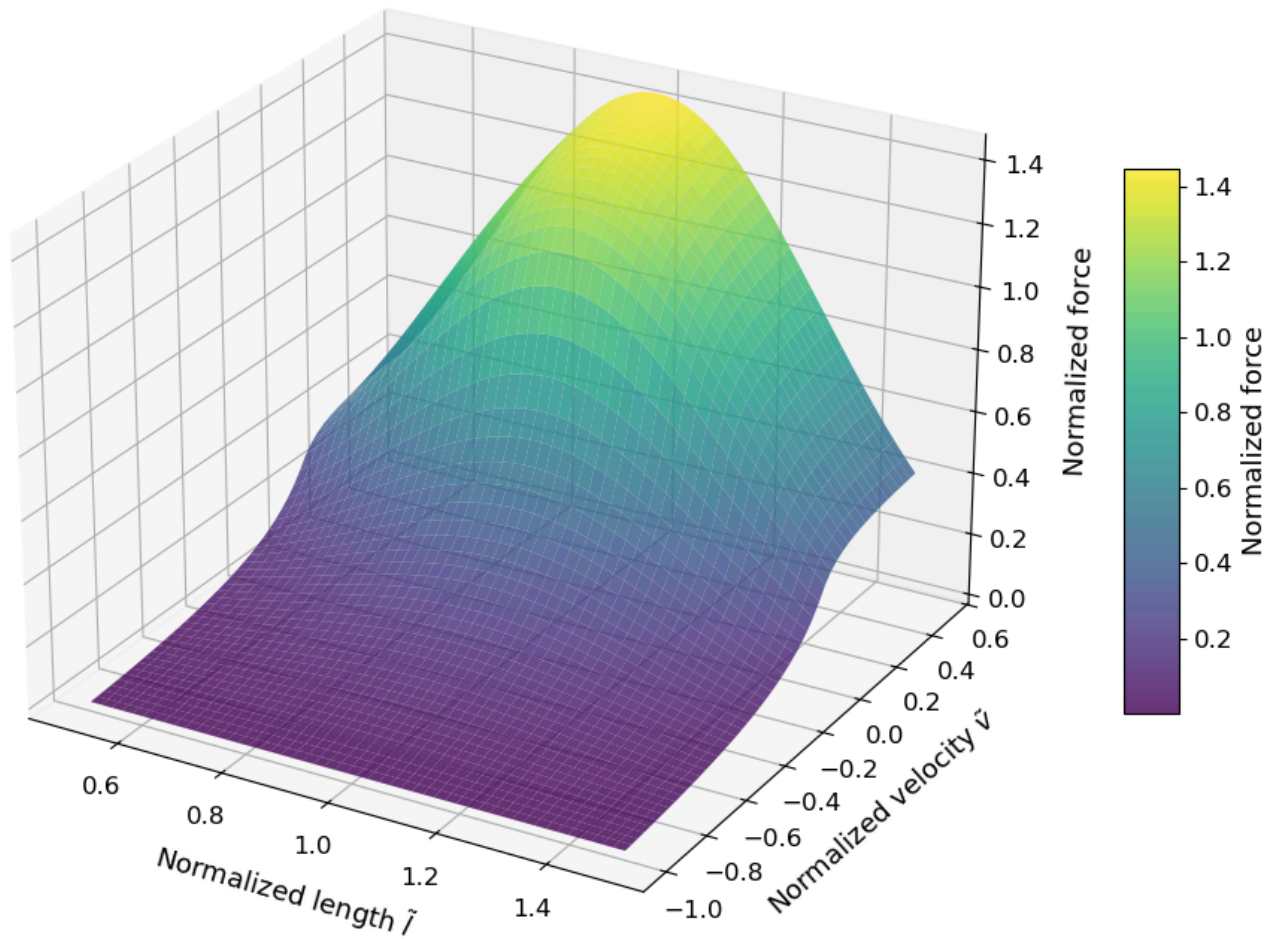
► Click to reveal answer

## 9. Python Implementation: Hill-Type Muscle Model

Now let's implement a complete Hill-type muscle model in Python. We'll build it step by step, starting with the individual components and then assembling them into a full simulation.

```
HillMuscleModel class defined successfully.  
Default parameters: F0=1000 N, l_opt=0.1 m, v_max=10 l_opt/s
```

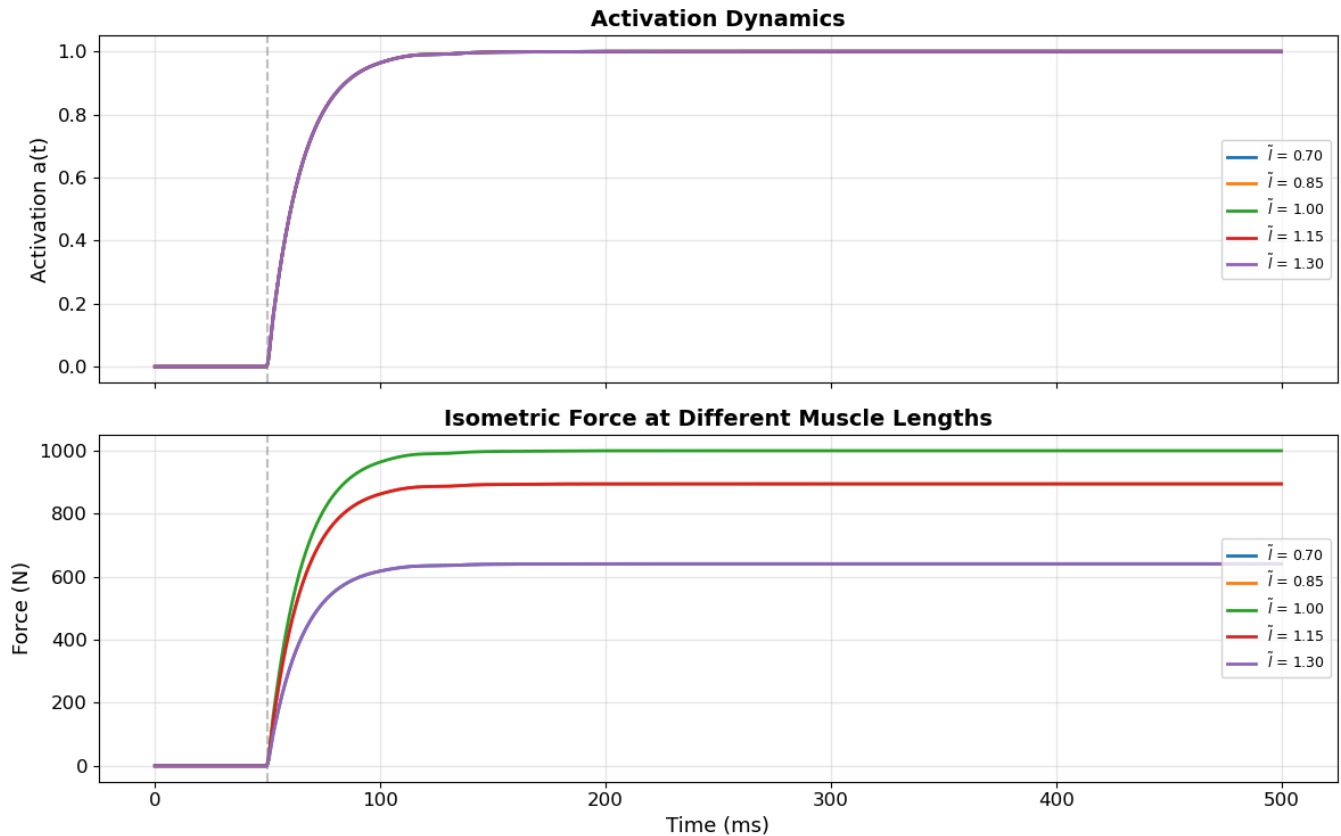
**Figure 8. Force-Length-Velocity Surface (Full Activation,  $a=1$ )**



## 9.1 Isometric Contraction Simulation

In an **isometric contraction**, the muscle length is held constant (velocity = 0) while activation increases. This is the simplest test of the model and mimics experiments where a muscle is stimulated while its tendons are fixed.

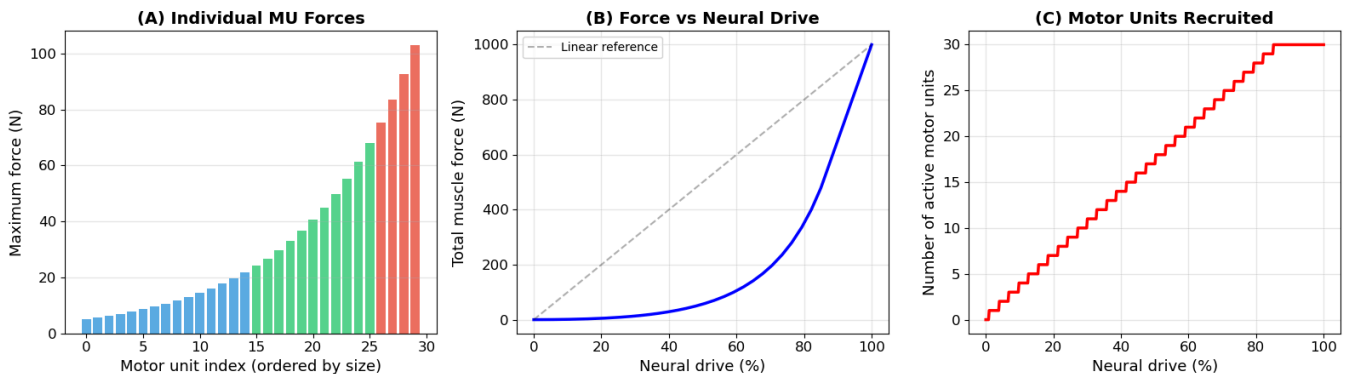
**Figure 9. Isometric Contraction Simulation**



## 9.2 Motor Unit Pool Simulation

Now let's simulate a more realistic scenario: a pool of motor units with different sizes, thresholds, and fiber types, following Henneman's size principle.

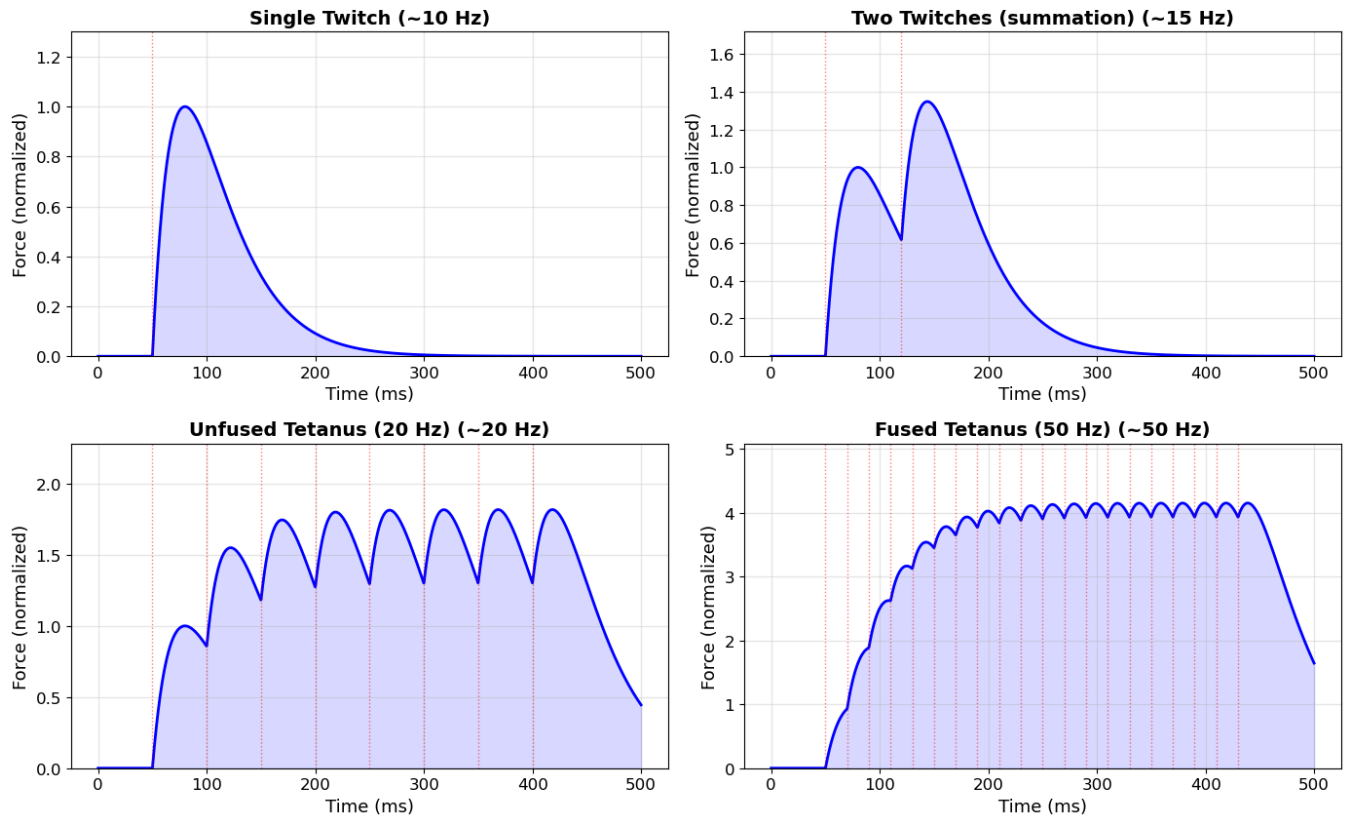
**Figure 10. Motor Unit Pool Simulation (30 Units, Size Principle)**



## 9.3 Muscle Twitch and Tetanus Simulation

A single action potential in a motor neuron produces a **twitch** — a brief, small contraction. When action potentials arrive in rapid succession, the twitches **summate** and at high enough frequencies produce a smooth, sustained **tetanus** — the maximum force the motor unit can generate.

**Figure 11. From Twitch to Tetanus: Temporal Summation of Muscle Force**



## Check Your Understanding: Twitches and Tetanus

**Q7.** Why does fused tetanus produce a higher force than a single twitch, even though each stimulus activates the same number of cross-bridges?

► Click to reveal answer

**Q8.** A patient with myasthenia gravis has reduced neuromuscular transmission efficiency. How would the force-frequency curve be affected?

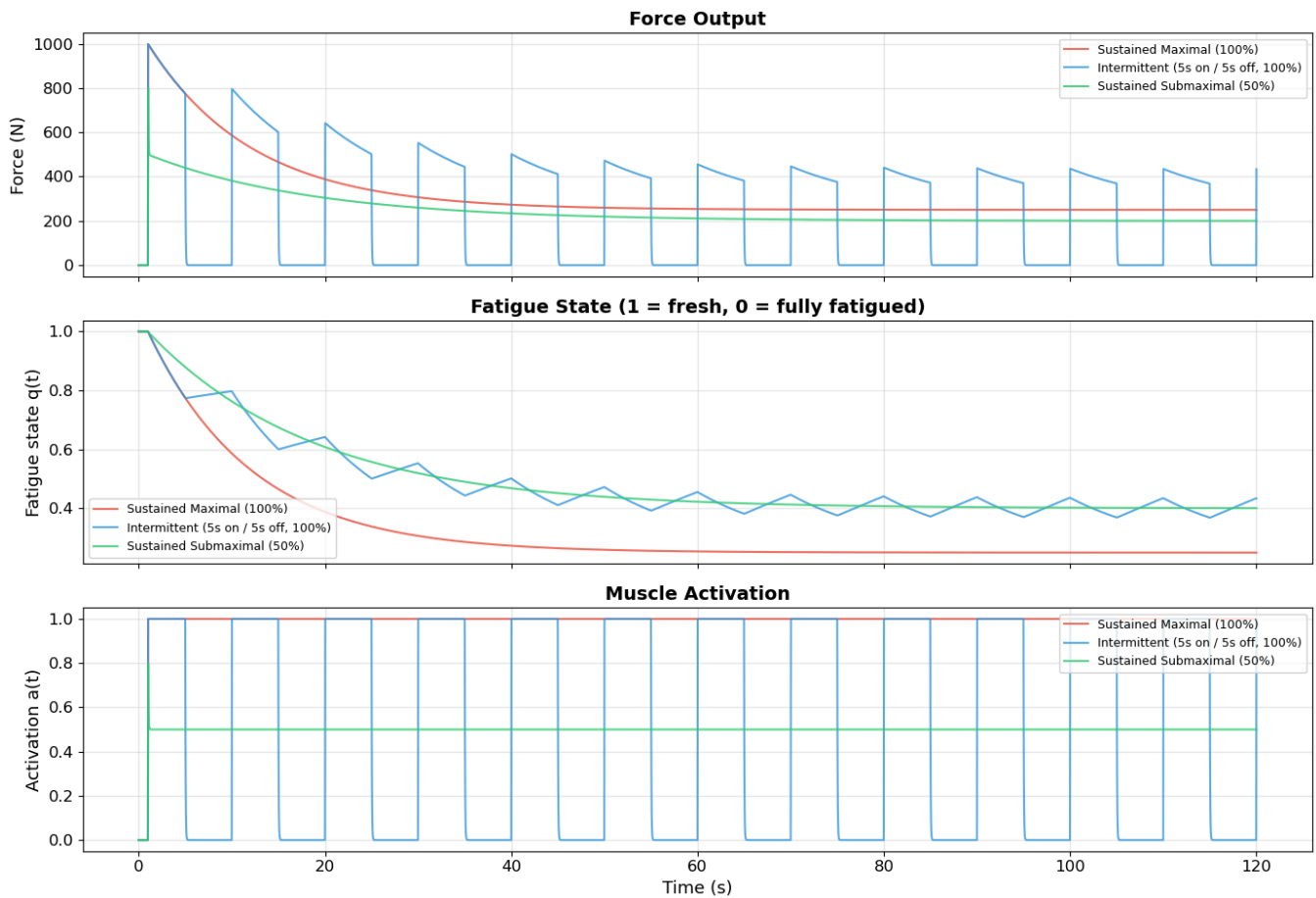
► Click to reveal answer

## 10. Muscle Fatigue Modeling

Real muscles fatigue — their force output declines with sustained or repeated contractions. Fatigue is a complex, multifactorial phenomenon involving both **central** (neural) and **peripheral** (muscular) mechanisms.

A simple phenomenological model captures the essential behavior:

**Figure 12. Muscle Fatigue Under Different Contraction Protocols**



## 11. Motor Control: From Neural Command to Movement

### The motor control hierarchy

Movement is controlled by a hierarchy of neural structures:

| Level              | Structure                              | Function                                            | Timescale  |
|--------------------|----------------------------------------|-----------------------------------------------------|------------|
| <b>Cortical</b>    | Motor cortex, premotor cortex, SMA     | Movement planning, voluntary commands               | 100–500 ms |
| <b>Subcortical</b> | Basal ganglia, cerebellum              | Action selection, error correction, timing          | 50–200 ms  |
| <b>Brainstem</b>   | Reticular formation, vestibular nuclei | Posture, balance, locomotion patterns               | 20–100 ms  |
| <b>Spinal</b>      | Motor neurons, interneurons, CPGs      | Motor unit recruitment, reflexes, rhythmic patterns | 1–50 ms    |
| <b>Peripheral</b>  | Muscles, tendons, joints               | Force generation, mechanical compliance             | 10–100 ms  |

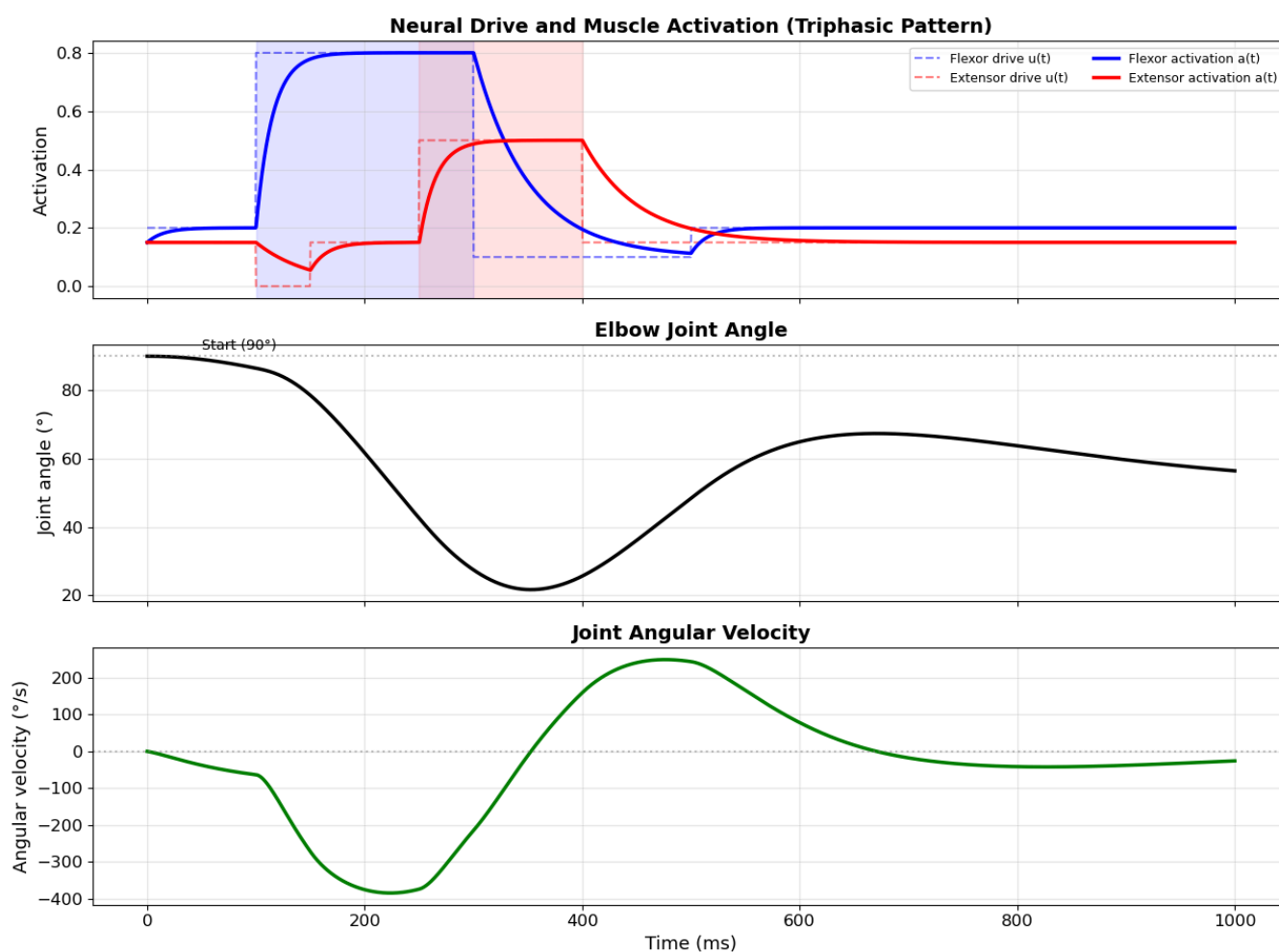
### Computational models of motor control

Modern theories of motor control are fundamentally computational:

1. **Internal models:** The brain maintains "forward models" (predict sensory consequences of actions) and "inverse models" (compute motor commands for desired outcomes)
2. **Optimal control:** Movements minimize a cost function (e.g. effort, jerk, endpoint variance)
3. **Feedback control:** Sensory feedback (proprioception, vision) is used to correct errors online
4. **Central Pattern Generators (CPGs):** Spinal circuits generate rhythmic motor patterns (e.g. walking) without continuous cortical input

Let's simulate a simple but instructive example: **controlling a single-joint limb with an agonist-antagonist muscle pair.**

**Figure 13. Simulated Reaching Movement with Agonist-Antagonist Muscles**



## The triphasic EMG pattern

The simulation above demonstrates a fundamental finding in motor control research: fast point-to-point movements are controlled by a **triphasic EMG pattern**:

1. **Agonist burst (AG1):** The prime mover (flexor) fires a strong burst to accelerate the limb toward the target

2. **Antagonist burst (ANT):** The opposing muscle (extensor) fires to decelerate the limb and prevent overshoot
3. **Second agonist burst (AG2):** A smaller agonist burst for final positioning and stabilization

This pattern is remarkably consistent across different movements, muscles, and species — suggesting it reflects a fundamental control strategy of the nervous system. It is also a key challenge in robotics: replicating this pattern with artificial actuators.

*The triphasic pattern is thought to be programmed by the central nervous system as a "motor program" — a pre-planned sequence that can be executed without relying on slow sensory feedback for the initial phases of rapid movement.*

## 12. From Humans to Robots: Bioinspired Motor Control

### Why robots need muscle-like actuators

Conventional robots use stiff, high-precision electric motors or hydraulic actuators. While excellent for industrial tasks, they struggle with:

- **Safety** around humans (rigid collisions cause injury)
- **Compliance** when interacting with soft, deformable objects
- **Energy efficiency** in dynamic tasks (walking, running)
- **Robustness** to unexpected perturbations

Biological muscles, by contrast, are inherently:

- **Compliant** — the force-length and force-velocity relationships provide natural "impedance"
- **Adaptive** — activation can be modulated continuously
- **Efficient** — elastic energy storage in tendons reduces metabolic cost
- **Robust** — redundant motor units provide graceful degradation

### Series Elastic Actuators (SEAs)

Inspired by the muscle-tendon unit, **Series Elastic Actuators** (SEAs) add a compliant element (spring) in series with a stiff motor. This mimics the role of the tendon and provides:

- Shock absorption
- Accurate force sensing (by measuring spring deflection)
- Energy storage and release (like the Achilles tendon during running)
- Safer human-robot interaction

SEAs are used in many modern humanoid robots, prosthetics, and exoskeletons.

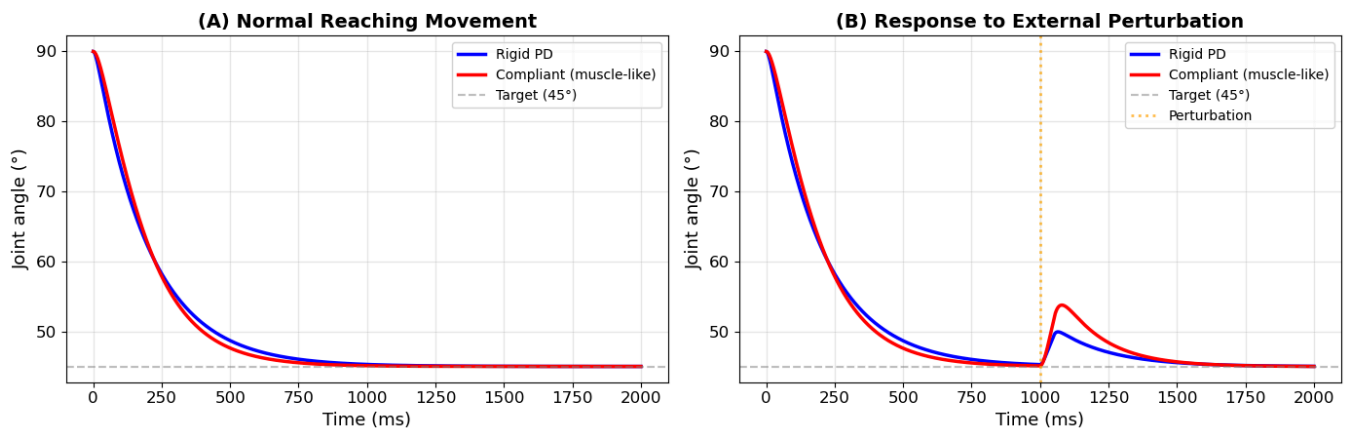
### Comparison: Biological muscle vs. robotic actuators



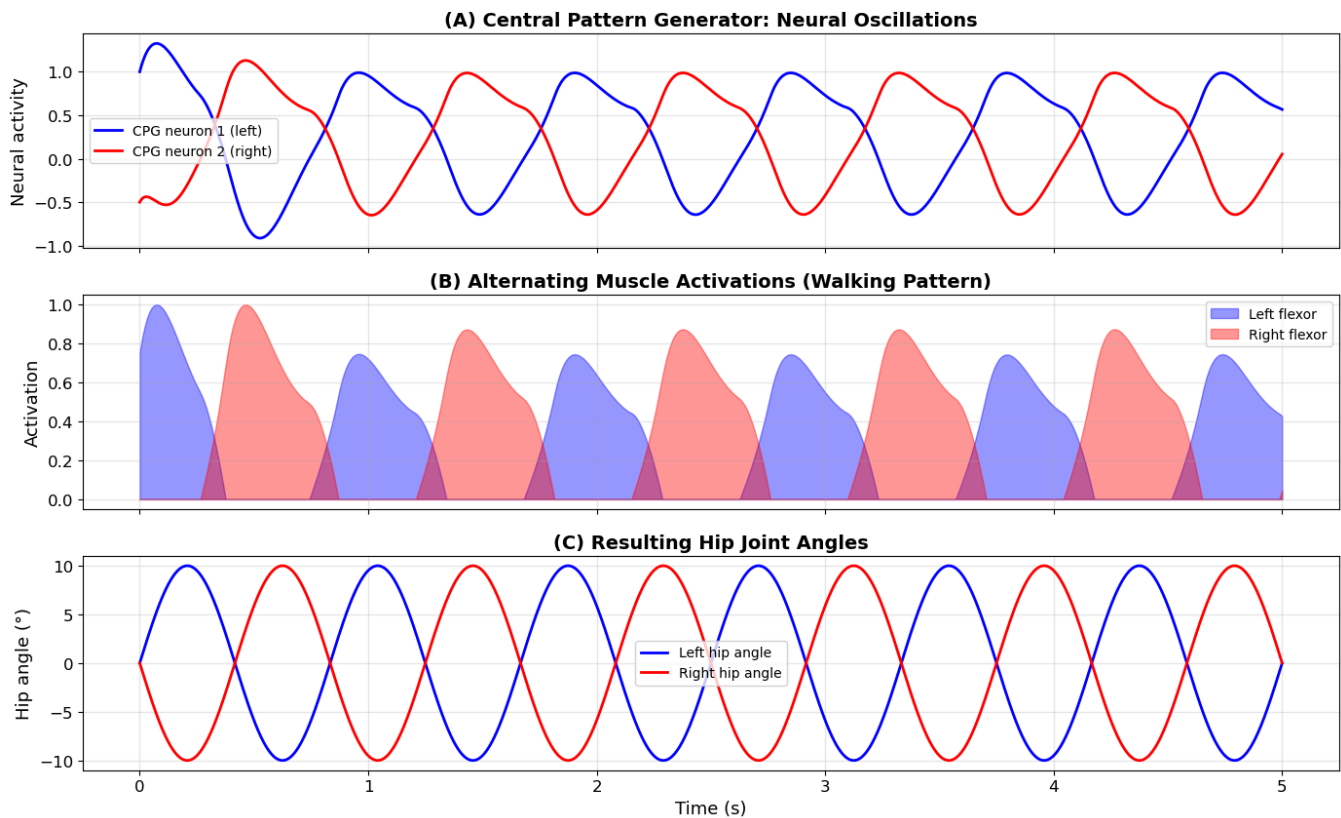
| Property              | Skeletal Muscle               | DC Motor      | SEA                   | Pneumatic Artificial Muscle |
|-----------------------|-------------------------------|---------------|-----------------------|-----------------------------|
| <b>Compliance</b>     | Intrinsic (F-L, F-V)          | None (rigid)  | Series spring         | Intrinsic (pressure-volume) |
| <b>Force control</b>  | Via recruitment + rate coding | Via current   | Via spring deflection | Via pressure                |
| <b>Power/weight</b>   | ~100–300 W/kg                 | ~100–500 W/kg | ~50–200 W/kg          | ~1000+ W/kg                 |
| <b>Bandwidth</b>      | 5–50 Hz                       | 100+ Hz       | 10–50 Hz              | 5–20 Hz                     |
| <b>Self-repair</b>    | Yes                           | No            | No                    | No                          |
| <b>Energy storage</b> | Tendon elasticity             | None          | Spring                | Minimal                     |

Let's simulate a simple robot arm using both rigid and compliant (muscle-like) control to see the difference.

**Figure 14. Rigid vs Muscle-Inspired Robot Control**



**Figure 15. Central Pattern Generator for Locomotion (Walking)**



## DIY Exercise: Explore the Simulations

**Exercise 3.** In the single-joint reaching simulation (Figure 13), what happens if you remove the antagonist brake entirely (set `u_extensor` to 0 during the movement)? Predict first, then modify the code.

► Click to reveal answer

**Exercise 4.** In the CPG simulation (Figure 15), what would happen if the mutual inhibition coupling strength is reduced to near zero? What does this model in terms of pathology?

► Click to reveal answer

**Exercise 5.** The robot comparison (Figure 14) shows that compliant control is more robust to perturbations. Can you think of a scenario where rigid control would be *preferable* to compliant control?

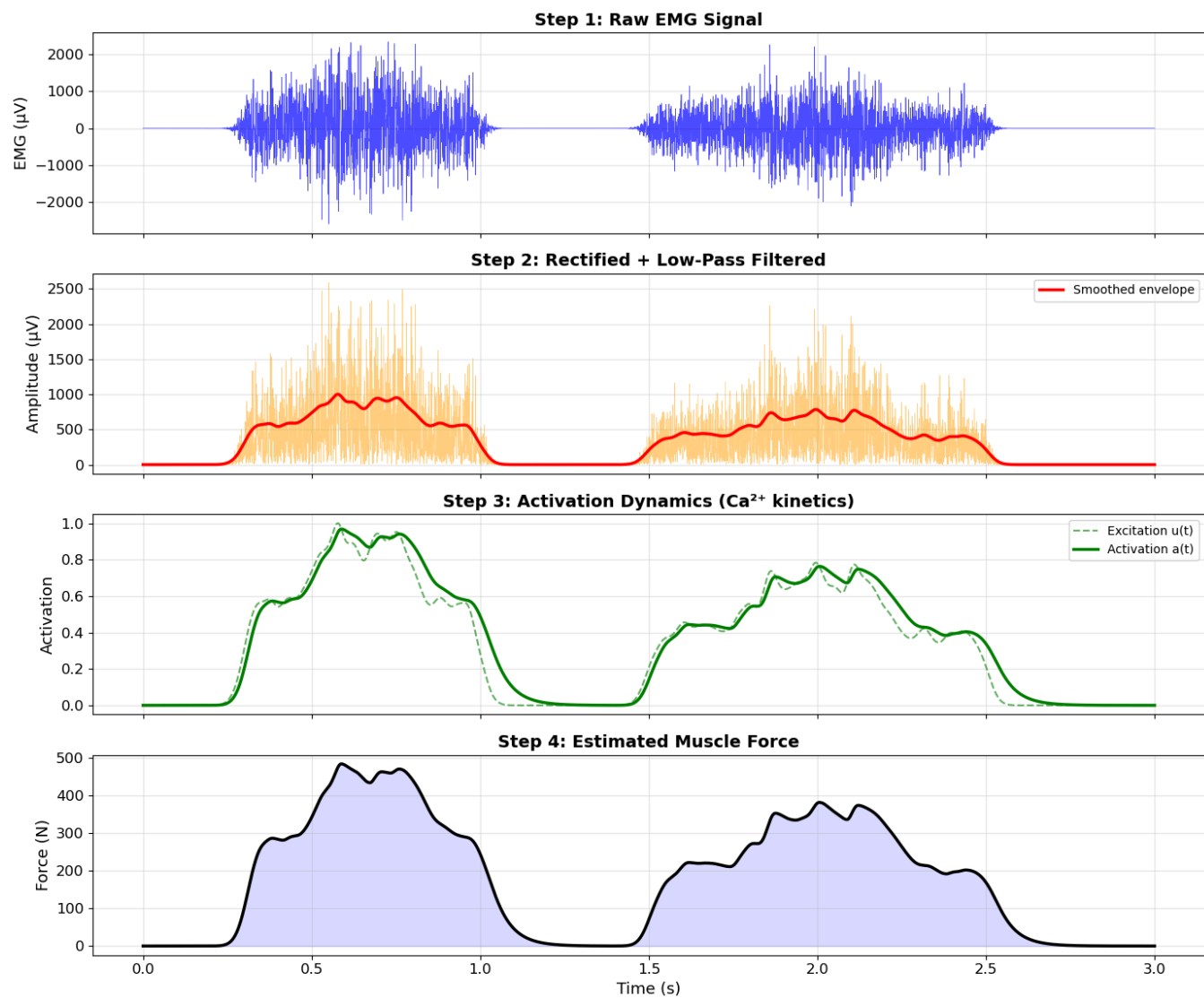
► Click to reveal answer

## 13. Putting It All Together: EMG-Driven Muscle Simulation

In clinical and research settings, we often record **electromyography (EMG)** signals — the electrical activity of muscles — and use computational models to estimate the forces those muscles produce. This is called **EMG-driven musculoskeletal modeling**.

Let's simulate a realistic scenario: processing a synthetic EMG signal through the full Hill-type model pipeline.

Figure 16. EMG-to-Force Pipeline: From Electrical Activity to Mechanical Force



# 14. Summary and Key Concepts

## What we've covered

| Topic                   | Key Insight                                                         |
|-------------------------|---------------------------------------------------------------------|
| Sliding filament theory | Contraction = filaments sliding past each other, not shortening     |
| Cross-bridge cycle      | Molecular motor powered by ATP hydrolysis (~5 nm, ~3 pN per stroke) |

| Topic                              | Key Insight                                                         |
|------------------------------------|---------------------------------------------------------------------|
| <b>Force-length relationship</b>   | Optimal length for maximum force (determined by filament overlap)   |
| <b>Force-velocity relationship</b> | Faster shortening → less force (Hill's hyperbolic equation, 1938)   |
| <b>Activation dynamics</b>         | Ca <sup>2+</sup> kinetics cause asymmetric activation/deactivation  |
| <b>Hill muscle model</b>           | CE + PEE + SEE = tractable model for simulation                     |
| <b>Motor unit recruitment</b>      | Size principle (Henneman): small → large, orderly                   |
| <b>Rate coding</b>                 | Higher firing rates → more force (temporal summation → tetanus)     |
| <b>Triphasic pattern</b>           | AG1-ANT-AG2 for fast, accurate point-to-point movements             |
| <b>Central Pattern Generators</b>  | Spinal circuits for rhythmic movements (walking)                    |
| <b>Fatigue</b>                     | Progressive force decline from metabolic and neural factors         |
| <b>Bioinspired robotics</b>        | Compliance, SEAs, and muscle-like control for safe, adaptive robots |

## The computational modeling perspective

The muscle system beautifully illustrates the power of **multi-scale modeling** in computational medicine:

- **Molecular scale:** Cross-bridge kinetics (A.F. Huxley, 1957)
- **Fiber scale:** Force-length, force-velocity (Hill, 1938)
- **Motor unit scale:** Recruitment and rate coding (Henneman, 1965)
- **Whole muscle scale:** Hill-type models (Zajac, 1989)
- **Musculoskeletal scale:** OpenSim, multi-body dynamics
- **Neural control scale:** Optimal control, CPGs, internal models
- **Robot design scale:** Bioinspired actuators and controllers

Each scale builds on the one below it, and computational models allow us to bridge these scales — from molecules to movement.

## Final DIY Challenges

**Challenge 1.** Modify the `HillMuscleModel` to add a **pennation angle** (the angle at which fibers are oriented relative to the tendon). In pennate muscles (like the gastrocnemius), fibers are at an angle  $\alpha$  to the tendon, so the effective force is  $F \cos(\alpha)$  but more fibers can be packed into a given volume. Implement this and show how pennation angle affects the force-length curve.

► Hint

**Challenge 2.** Create a **two-joint arm model** (shoulder + elbow) controlled by four muscles (shoulder flexor/extensor + elbow flexor/extensor). Simulate a reaching movement to a target in 2D space using a simple feedback controller.

► Hint

**Challenge 3.** Implement a **muscle fatigue model with fiber-type-specific fatigue rates**. Type I fibers should fatigue slowly while Type IIx fibers fatigue rapidly. Simulate a sustained 60% MVC contraction and show how the motor unit pool must progressively recruit larger (and more fatigable) units to maintain the target force — a phenomenon known as **muscle wisdom** or **rotation of motor units**.

► Hint

# 15. Robot Design and Bioinspired Actuators: From Muscle to Machine

## Why biology is the best robotics teacher

The field of robotics is undergoing a paradigm shift. For decades, robots were designed around the principles of **rigid-body mechanics**: stiff links, high-precision gearboxes, and position-controlled electric motors. While these machines excel in structured factory environments, they fundamentally struggle in the unstructured, dynamic world that humans and animals navigate effortlessly.

The reason is profound: **biology solves motor control differently from engineering**. Muscles are not motors. Tendons are not rigid linkages. The spinal cord is not a PID controller. By understanding *how* biology solves movement, we can design better robots — and conversely, building robots that mimic biology deepens our understanding of how the body works.

This section explores the rich intersection of muscle physiology, control theory, and modern robotics.

### 15.1 The limitations of conventional robotics

Traditional industrial robots (e.g., KUKA, ABB, Fanuc) use:

- **Stiff actuators:** High gear-ratio motors that resist perturbations with brute force
- **Position control:** Precise trajectory tracking via PID or computed torque control
- **Rigid structure:** Minimal compliance in links and joints

This works well for repetitive tasks in caged environments, but fails when:

| Challenge       | Why conventional robots struggle               | How biology solves it                                              |
|-----------------|------------------------------------------------|--------------------------------------------------------------------|
| Human proximity | Rigid collisions cause injury                  | Muscles are inherently compliant; impedance modulates with context |
| Uneven terrain  | Precise trajectory control assumes flat ground | Reflexes and CPGs adapt gait in real-time                          |

| Challenge                           | Why conventional robots struggle                   | How biology solves it                                                                                 |
|-------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Manipulating fragile objects</b> | Force control bandwidth is limited by sensor delay | Muscle's force-velocity relationship provides instant "damping"                                       |
| <b>Energy efficiency</b>            | Motors waste energy fighting gravity and inertia   | Tendons store and release elastic energy (e.g., Achilles tendon saves ~35% metabolic cost in running) |
| <b>Robustness to damage</b>         | Single motor failure = system failure              | Redundant motor units degrade gracefully (Henneman's size principle)                                  |

## 15.2 Bioinspired actuator technologies

Inspired by muscle, engineers have developed several classes of **soft and compliant actuators**:

### Series Elastic Actuators (SEAs)

- **Concept:** A stiff motor in series with a compliant spring (mimicking the muscle-tendon unit)
- **Advantages:** Force sensing via spring deflection, shock absorption, energy storage
- **Examples:** MIT Cheetah robot, NASA Valkyrie, Agility Robotics Digit
- **Biological analog:** The Hill muscle model's series elastic element (SE) is precisely the engineering inspiration

### Variable Impedance Actuators (VIAs)

- **Concept:** Actuators whose stiffness can be *actively modulated* — just as muscles co-contract to stiffen a joint
- **Types:** Variable stiffness actuators (VSAs), antagonistic actuators (two SEAs pulling in opposite directions)
- **Biological analog:** Agonist-antagonist muscle pairs with co-contraction for impedance control
- **Key insight:** The brain doesn't just control position and force — it controls **impedance** (the ratio of force to displacement). VIAs can too.

### Pneumatic Artificial Muscles (PAMs / McKibben muscles)

- **Concept:** A braided mesh over an inflatable bladder; pressurizing shortens the actuator (like muscle contraction)
- **Advantages:** Extremely high power-to-weight ratio (~1000 W/kg, vs ~300 W/kg for skeletal muscle), inherent compliance
- **Disadvantages:** Require compressed air, limited bandwidth (~5–20 Hz), hysteresis
- **Biological analog:** Direct mimic of whole-muscle contraction
- **Used in:** Shadow Dexterous Hand, Festo BionicCobot, exoskeletons for rehabilitation

### Dielectric Elastomer Actuators (DEAs)

- **Concept:** An elastomer membrane sandwiched between compliant electrodes; applying voltage causes area expansion

- **Advantages:** Lightweight, silent, flexible, large strain (>100%)
- **Disadvantages:** Require high voltage (~kV), low force density
- **Biological analog:** Mimics the deformability and distributed nature of muscle tissue
- **Used in:** Soft grippers, wearable haptics, microrobots

### Shape Memory Alloys (SMAs) and Twisted Polymer Fibers

- **Concept:** Materials that change shape with temperature (SMA) or twist under heating (polymer fibers)
- **Advantages:** Extremely compact, no external mechanism
- **Disadvantages:** Slow (thermal cycling), limited cycle life (SMA), low efficiency
- **Recent advances (2025):** Fiber-type artificial muscles combining SMA and polymer fibers achieve muscle-like contraction ratios in sub-millimeter form factors ([Hwang et al., Nature Reviews Materials, 2025](#))

### Hydraulic Artificial Muscles

- **Concept:** Fluid-driven actuators with high force and speed
- **Examples:** Boston Dynamics’ original hydraulic Atlas robot (retired 2024)
- **Transition:** The 2024 Electric Atlas replaced hydraulics with electric actuators for quieter, cleaner operation — reflecting a broader industry shift

## 15.3 Comprehensive comparison: biological muscle vs. engineered actuators

| Property       | Skeletal Muscle             | DC Motor + Gearbox  | SEA               | McKibben (PAM)       | DEA                   | SMA Fiber   |
|----------------|-----------------------------|---------------------|-------------------|----------------------|-----------------------|-------------|
| Compliance     | Intrinsic (F-L, F-V curves) | None (rigid)        | Tunable (spring)  | Intrinsic (pressure) | Intrinsic (elastomer) | Minimal     |
| Force control  | Recruitment + rate coding   | Current control     | Spring deflection | Pressure regulation  | Voltage control       | Temperature |
| Power/weight   | ~100–300 W/kg               | ~100–500 W/kg       | ~50–200 W/kg      | ~1000+ W/kg          | ~10–50 W/kg           | ~50 W/kg    |
| Bandwidth      | 5–50 Hz                     | 100+ Hz             | 10–50 Hz          | 5–20 Hz              | 10–100 Hz             | 0.1–1 Hz    |
| Max strain     | ~40%                        | Continuous rotation | Depends on spring | ~25–35%              | >100%                 | ~5–8%       |
| Self-repair    | Yes (regeneration)          | No                  | No                | No                   | Limited               | No          |
| Energy storage | Tendon elasticity           | None                | Spring            | Minimal              | Dielectric            | Lattice     |
| Efficiency     | ~25% (mechanical)           | ~80–95%             | ~60–80%           | ~20–30%              | ~10–30%               | ~3–5%       |
| Noise          | Silent                      | Motor hum           | Quiet             | Compressor noise     | Silent                | Silent      |

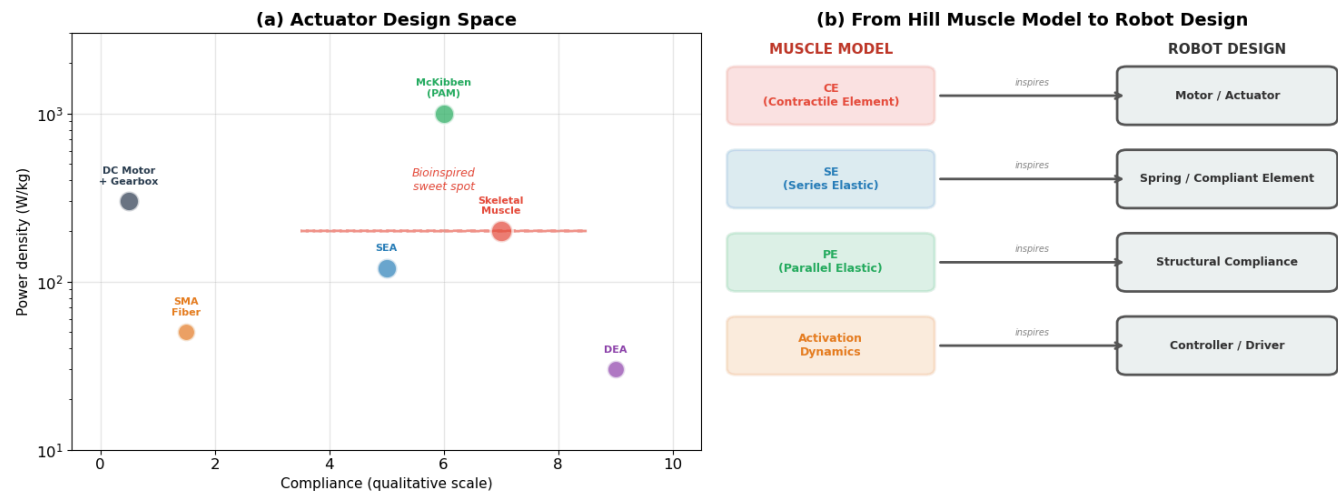
# 15.4 Modern humanoid robots: the state of the art (2024–2025)

The humanoid robot landscape has entered a remarkable era of acceleration:

| Robot             | Organization     | Actuator Type                            | Key Bioinspired Feature                                                     |
|-------------------|------------------|------------------------------------------|-----------------------------------------------------------------------------|
| Atlas (Electric)  | Boston Dynamics  | Custom electric with SEA-like compliance | Whole-body dynamic balance; parkour-level agility                           |
| Optimus (Gen 2)   | Tesla            | Electric with N+1 DOF hands              | Neural-network-based control; learning from demonstration                   |
| Digit             | Agility Robotics | SEAs in legs                             | Compliant walking on uneven terrain; warehouse logistics                    |
| Figure 02         | Figure AI        | Electric                                 | LLM-integrated planning; OpenAI partnership                                 |
| H1 / G1           | Unitree          | Quasi-direct-drive motors                | Low-cost, high-torque; reinforcement-learning-trained locomotion            |
| Sanctuary Phoenix | Sanctuary AI     | Electric + tendon-driven hands           | World’s most dexterous robot hand (20 DOF, tendon-actuated like human hand) |
| LIMS2-AMBIDEX     | KAIST            | Tendon-driven arms                       | Cable-driven antagonistic actuation mimicking human arm muscles             |

**Key trend (2024–2025):** The industry is converging on **learning-based control** (reinforcement learning, imitation learning) combined with **compliant hardware** — precisely the combination that biology uses (spinal learning + compliant muscles). The Hill muscle model properties we studied above (force-length, force-velocity, impedance modulation) are the *design targets* these engineers are pursuing.

Figure 17. Bioinspired Actuator Design: From Muscle Physiology to Robot Engineering



# 15.5 Bioinspired controllers: how robots learn from the spinal cord



Just as bioinspired *actuators* mimic muscle properties, bioinspired *controllers* mimic the neural circuits that drive movement. The key insight from motor neuroscience is that locomotion is largely controlled by **spinal circuits** — not the brain:

## Central Pattern Generators (CPGs)

- **Biology:** Networks of spinal interneurons that generate rhythmic motor patterns (walking, swimming, breathing) *without* sensory input or cortical commands
- **Key property:** CPGs produce basic rhythmic patterns that are *modulated* (not generated) by higher-level commands and sensory feedback
- **Robot application:** CPG models (often based on coupled oscillators, as in Figure 15 above) are the dominant paradigm for legged robot locomotion
- **Recent advances (2025):** Bio-inspired neural networks that combine CPGs with multi-layer perceptrons enable quadruped robots to traverse uneven terrain, resist perturbations, and execute multiple gaits within a single control policy ([Nachstedt et al., Scientific Reports, 2025](#))

## Reflex-based control

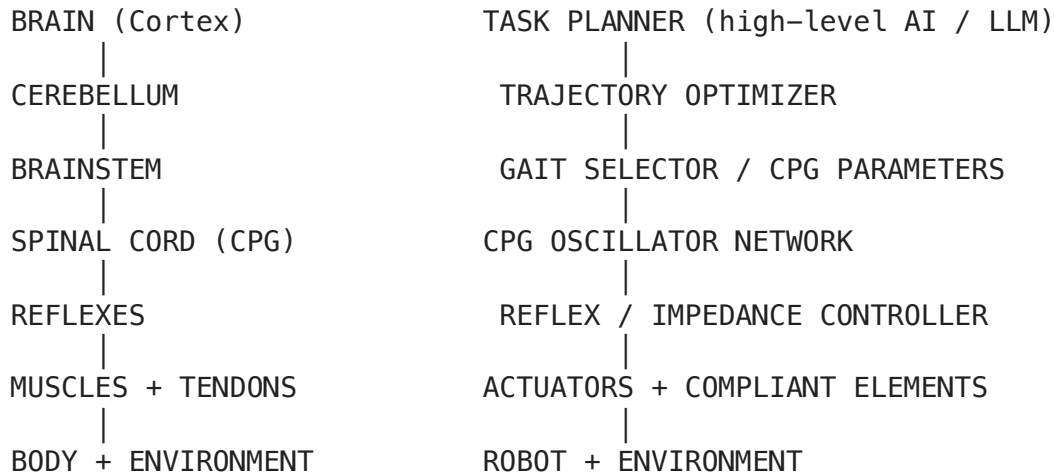
- **Biology:** Proprioceptive reflexes (stretch reflex, Golgi tendon reflex) modulate muscle activation based on real-time sensory feedback from muscle spindles and Golgi tendon organs
- **Key property:** Reflexes provide *local, fast* corrections without waiting for brain commands (~30 ms vs ~200 ms)
- **Robot application:** Virtual reflexes in robot controllers adjust joint torques based on force/position sensor feedback
- **Recent advances (2025):** Taga et al.'s work has been extended to show that proprioceptive reflex pathways *alone* — without CPG rhythm generators — can produce diverse bipedal gaits including walking, hopping, and running ([Van der Noot et al., Communications Biology, 2025](#))

## The impedance control paradigm

- **Biology:** The nervous system doesn't just control position or force — it controls the *relationship* between them (impedance). By modulating muscle co-contraction, the brain adjusts joint stiffness and damping to the task:
  - **High stiffness:** Holding a heavy object steady
  - **Low stiffness:** Drawing with a pen (allowing compliance for smooth curves)
  - **Variable stiffness:** Catching a ball (stiff at impact, then compliant to absorb)
- **Robot application: Impedance control** (Hogan, 1985) explicitly controls the dynamic relationship between robot position and force, rather than either alone. This is now standard in collaborative robots (cobots)
- **Mathematical connection:** The Hill model's force-velocity relationship ( $F \propto -v$  near isometric) is precisely a **viscous damping** term; the force-length relationship ( $F \propto -\Delta L$  near optimal length) is a **spring-like restoring force**. Together, they define the muscle's **impedance** — the same quantity robot controllers modulate

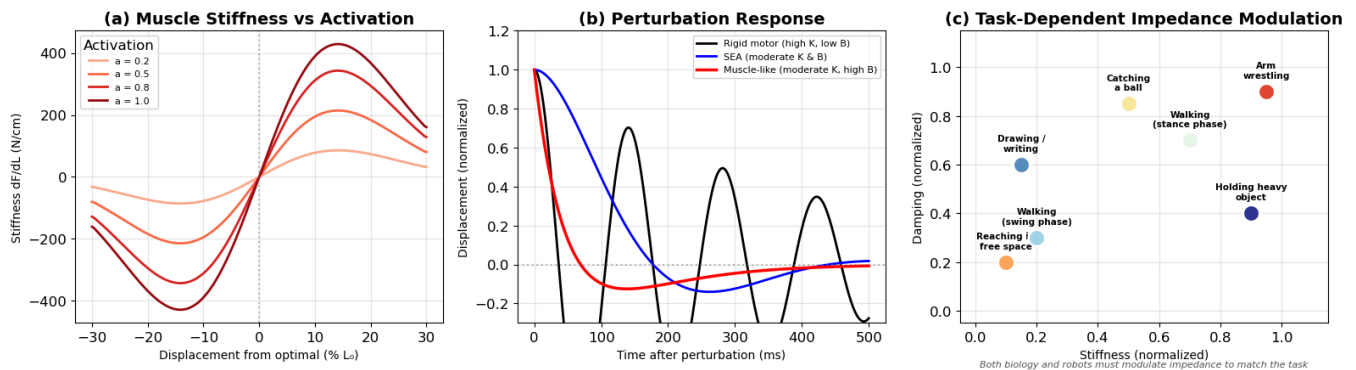
## Hierarchical motor control in robots

Modern bioinspired robot architectures mirror the biological motor hierarchy:



At every level, the biological and robotic architectures face the same fundamental challenges: **coordination** (syncing many degrees of freedom), **adaptation** (adjusting to changing conditions), and **robustness** (handling unexpected perturbations).

Figure 18. Impedance as a Unifying Principle: Muscle Physiology → Robot Control



## Check Your Understanding: Robot Design and Bioinspired Actuators

**Q10.** Why do Series Elastic Actuators (SEAs) perform better than rigid motors when a robot must interact physically with humans (e.g., collaborative manufacturing, rehabilitation)?

► Click to reveal answer

**Q11.** A rehabilitation exoskeleton needs to assist a patient with weakened leg muscles during walking. Which actuator type(s) would be most appropriate, and why?

► Click to reveal answer

**Q12.** In Figure 18(b), the muscle-like system returns to equilibrium more smoothly than the rigid motor. What property of the Hill model produces this superior damping?

► Click to reveal answer

---

**Q13.** Looking at Figure 18(c), why does “catching a ball” require BOTH high stiffness and high damping, while “drawing” requires low stiffness and moderate damping?

► Click to reveal answer

## 15.6 The frontier: where muscle science meets AI

The most exciting current developments sit at the intersection of muscle biomechanics, soft robotics, and machine learning:

### Reinforcement learning on compliant hardware

Companies like Agility Robotics, Unitree, and Boston Dynamics now train locomotion policies using **reinforcement learning** (RL) in simulation, then deploy them on compliant robot hardware. The compliant actuators provide the mechanical robustness that allows RL policies — which are imperfect — to work in the real world. This is analogous to how babies learn to walk: the intrinsic compliance of immature muscles provides a safety margin while the neural controller is still learning.

### Muscle-level digital twins

The Hill muscle model and its extensions are now embedded in **digital twin** platforms (e.g., OpenSim, AnyBody) that create personalized musculoskeletal models of individual patients. These are used to:

- Plan orthopedic surgeries (e.g., tendon transfer, joint replacement)
- Design custom prosthetics and exoskeletons
- Optimize rehabilitation protocols
- Predict post-surgical outcomes

### Biohybrid robots

The most literal form of bioinspired actuators: **biohybrid robots** use *living muscle tissue* grown on engineered scaffolds as actuators. Recent demonstrations include:

- Swimming robots powered by rat cardiomyocytes (Nawroth et al., 2012 — the “medusoid”)
- Walking biobots driven by skeletal muscle tissue optogenetically stimulated by light (Raman et al., 2016)
- Finger-like grippers actuated by living muscle (Morimoto et al., 2018)

These blur the boundary between organism and machine, and represent a future where the muscle models in this notebook are not just simulations — but actual engineering design tools for living actuators.

## DIY Exercise: Design Your Own Bioinspired Robot

**Exercise 5.** You are designing a robot arm for a nursing home that must:

- Hand objects to elderly patients
- Assist with standing up from a chair
- React safely to unexpected contact

Based on what you've learned in this notebook, specify:

1. What actuator type(s) would you use and why?
2. What control architecture (position control, force control, impedance control)?
3. Which properties of the Hill muscle model are most important to replicate?

► Click to reveal a suggested design

## 16. Going Deeper

### Recommended reading

#### Foundational papers:

- Hill, A.V. (1938). "The heat of shortening and the dynamic constants of muscle." *Proceedings of the Royal Society B*, 126(843), 136–195.
- Huxley, A.F. & Niedergerke, R. (1954). "Structural changes in muscle during contraction." *Nature*, 173, 971–973.
- Huxley, H.E. & Hanson, J. (1954). "Changes in the cross-striations of muscle during contraction." *Nature*, 173, 973–976.
- Huxley, A.F. (1957). "Muscle structure and theories of contraction." *Progress in Biophysics and Biophysical Chemistry*, 7, 255–318.
- Henneman, E., Somjen, G., & Carpenter, D.O. (1965). "Functional significance of cell size in spinal motoneurons." *Journal of Neurophysiology*, 28(3), 560–580.
- Zajac, F.E. (1989). "Muscle and tendon: properties, models, scaling, and application to biomechanics and motor control." *Critical Reviews in Biomedical Engineering*, 17(4), 359–411.

#### Textbooks:

- Nigg, B.M. & Herzog, W. (2007). *Biomechanics of the Musculo-skeletal System*. Wiley.
- Winter, D.A. (2009). *Biomechanics and Motor Control of Human Movement*. Wiley.
- Kandel, E.R. et al. (2021). *Principles of Neural Science*. McGraw-Hill. (Chapters on motor systems)

#### Bioinspired robotics and actuators:

- Pratt, G.A. & Williamson, M.M. (1995). "Series elastic actuators." *IEEE/RSJ Int. Conf. on Intelligent Robots and Systems*. — The foundational SEA paper

- Hogan, N. (1985). "Impedance control: An approach to manipulation." *J. Dynamic Systems, Measurement, and Control*, 107, 1–24. — The foundational impedance control paper
- Ijspeert, A.J. (2008). "Central pattern generators for locomotion control in animals and robots." *Neural Networks*, 21(4), 642–653.
- Nachstedt et al. (2025). "Bio-inspired neural networks with CPGs for learning multi-skill locomotion." *Scientific Reports*. — CPG + RL for quadruped locomotion
- Van der Noot et al. (2025). "A spinal network of proprioceptive reflexes can produce a variety of bipedal gaits." *Communications Biology*.
- Hwang et al. (2025). "Fiber-type artificial muscles for robotic actuation." *Nature Reviews Materials*. — SOTA review of artificial muscle fibers

### Software and tools:

- [OpenSim](#) — Open-source musculoskeletal modeling and simulation
- [PyBullet](#) — Physics simulation for robotics
- [MuJoCo](#) — Multi-Joint dynamics with Contact (now open source)

## Extensions to explore

1. **Detailed cross-bridge models:** Replace the Hill-type model with Huxley's 1957 cross-bridge model (PDE-based)
2. **Multi-muscle, multi-joint simulations:** Use OpenSim to simulate realistic musculoskeletal systems
3. **Optimal control of movement:** Implement minimum-jerk or minimum-effort trajectory optimization
4. **Machine learning for motor control:** Train neural networks to control simulated musculoskeletal systems
5. **Cardiac muscle modeling:** Apply cross-bridge theory to model heart contraction
6. **Smooth muscle:** Model the very different contraction dynamics of vascular and visceral smooth muscle
7. **Variable impedance control:** Implement a robot controller that modulates stiffness and damping based on task context
8. **Sim-to-real transfer:** Train a RL locomotion policy in PyBullet/MuJoCo with Hill-model-based actuators, then deploy on real hardware
9. **Biohybrid actuation:** Model living muscle tissue on engineered scaffolds — what parameters of the Hill model change?

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*This notebook is part of the BMED365 Computational Medicine course at the University of Bergen. The models presented here are simplified for pedagogical purposes — real musculoskeletal models (e.g. those in OpenSim) include many additional features such as pennation angles, tendon dynamics, muscle wrapping around bones, and detailed joint kinematics.*